Linköping University | Department of Computer and Information Science
Master's thesis, 30 ECTS | Cognitive Science
Spring 2026 | LIU-IDA/KOGVET-A–26/005–SE

Designing Cobot-to-Human Communication

– A Cognitive Ergonomic Approach to Industrial Human-Robot Collaboration

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Abstract

Collaborative robots (cobots) are of increasing use in industry, and transparency has been identified as an important aspect of cobot implementation. Transparency includes the cobot communicating information about its states, goals, processes, and so forth to the worker. Presenting this information intuitively can aid in solving prominent cognitive ergonomic issues that can arise in human-robot collaboration. The aim of this thesis was twofold, to develop design concepts for cobot-to-human communication with stakeholder involvement, and to evaluate developed design concepts in terms of cognitive ergonomic and user experience outcomes. Design development included concept generation through workshops with university students, and expert evaluation of generated concepts through an online feedback questionnaire. Two of the concepts were further developed, one for an independent scenario (human and cobot doing different tasks on different workpieces) which communicated cobot states through salient colour signals. The other concept was for a simultaneous scenario (human and cobot doing different tasks but on the same workpiece), where the cobot's state was communicated through an integrated cobot screen and upcoming movement through a light band around the cobot's arm. These were each empirically assessed in comparison to a baseline through a within-subjects questionnaire administered to industrial workers. Results showed that the developed concept for the independent scenario was better in terms of situation awareness, perceived safety, and user experience but not different to baseline on cognitive workload. For the simultaneous scenario, there was no difference between the developed design and baseline on any outcome. The conclusion was that the cognitive ergonomics and user experience of workers can be improved through basic signalling in an independent scenario, but more transparency information or improved interface design may be needed for simultaneous scenarios.

Key Words: *collaborative robots, cognitive ergonomics, robot communication, perceived safety, user experience, industrial human-robot collaboration*

Acknowledgments

I would like to express my appreciation to my supervisors, Jennifer Renoux and Franziska Babel, for their guidance, input and help throughout the thesis process. I also want to give the sincerest of thanks to our contacts at partnering industrial companies, who participated in the study and helped organise access to industrial workers for data collection. Without your help, this thesis would not have been possible. Also a big thanks to my classmates, who during seminars gave valuable insight to the theoretical grounding of methodology as well as improved structure of the thesis. Finally, to friends and family who have listened, supported, and believed in me along the way.

Statement on Generative AI

Generative AI has been used in this thesis to improve the quality of language (e.g. spelling, grammar, and flow) and to check for inconsistencies (e.g. acronyms, figure referrals). It was also used to transfer thesis elements into LaTeX (e.g. tables/figures) and learn LaTeX functions (e.g. referrals to appendices and figures). For more information see Appendix 11.8.

Generative AI has not been used to create text content (such as ideas, arguments, or references).

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1 Introduction

Collaborative robots (cobots) are a central technology in industrial human-robot collaboration (HRC). Advancing cobot technology was particularly pertinent during the fourth industrial revolution (Industry 4.0), where the focus was on digitalisation and how it shapes industrial processes and changes worker roles (Davies, 2015; European Commission: Directorate-General for Research and Innovation, 2020). With the fifth industrial revolution in 2020 (Industry 5.0), focus expanded beyond digitalisation technology to include resilient, sustainable, and human-centric perspectives (Alves et al., 2023; European Commission: Directorate-General for Research and Innovation, 2020, 2024). In terms of human-centricity, Industry 5.0 emphasises worker empowerment, satisfaction, and well-being (European Commission: Directorate-General for Research and Innovation, 2020).

This shift is particularly relevant for cobot applications as cobots share workspaces and tasks with humans, meaning that not only safety but also the workers' experience of the interaction is an important aspect of their implementation. Accordingly, it is central to consider workers' user experience (UX) and cognitive ergonomics in industrial HRC. Previous research largely neglected these aspects, but since Industry 5.0, more cobot research has considered both cognitive ergonomics (Attahir et al., 2025; Berti et al., 2025; Cardoso et al., 2021; Concetta Manuela et al., 2025; Gualtieri et al., 2021, 2022; Lorenzini et al., 2023), and UX (Fournier et al., 2022; Ioniță et al., 2025; Nielsen et al., 2026). Cognitive ergonomics concerns optimising human well-being and system performance through knowledge of mental processes related to interactions between humans and other elements of a system (Karwowski & Zhang, 2021). Previous research on cognitive ergonomics in human-robot interaction (HRI) identifies situation awareness (SA) and cognitive workload as two central issues (Bassi et al., 2025; Chen & Barnes, 2021; Ioniță et al., 2025; Villani et al., 2018).

In cobot settings, the cobot often works semi-autonomously which inherently reduces the SA of the worker (Endsley, 1995; Endsley, 2021), meaning workers must acquire much of the SA on demand. To support SA in such contexts, it is important that relevant information is communicated to the workers (Cai & Patrick Rau, 2026; Chen et al., 2014). In the context of cobots, relevant information for the worker includes, for example, the cobot's state, task progress, and movement intentions (Dei et al., 2025; Gross & Krenn, 2024; Hoecherl et al., 2018; Scholz et al., 2025). Communicating such information makes the cobot more transparent to the worker (Chen et al., 2014), which has been shown to be beneficial for cognitive and UX

factors such as perceived safety (Vermeulen et al., 2024), SA (Cai & Patrick Rau, 2026), and cognitive workload (Cai & Patrick Rau, 2026; Carissoli et al., 2024).

At the same time, this information needs to be presented in a way that ensures the worker's cognitive workload is not negatively affected by increased information intake (Chen & Barnes, 2021; Chen et al., 2014). This is especially relevant since HRC often involves task and attention switching for the worker, which can strain cognitive workload (Chen & Barnes, 2021). Industrial cobot contexts are also noisy in terms of both auditory and visual input, which can further affect cognitive workload. This also requires signals to be clear and attention-grabbing.

Overall, robot-to-human (RtH) communication needs to support SA, perceived safety, and UX, while not increasing workload. This highlights the importance of well-designed communication interfaces. Useful approaches to reach these aims include cognitive science and participatory design. The cognitive science perspective can aid the design process by considering cognitive constraints as described above. It can also support the subsequent testing of designs to ensure they do not negatively affect relevant cognitive factors. Participatory design involves engaging stakeholders in the design process, which is important in understanding how to support UX, but also to generally get expert input on design concepts early in the design process.

Despite a growing attention to RtH communication interfaces, several gaps exist in the cobot-specific literature in relation to stakeholder involvement, testing designs with industrial workers, and the industrial scenarios investigated. First, previous cobot communication studies rarely involved stakeholders in the design process, despite it being identified as needed and valuable (Cao et al., 2024; Nielsen et al., 2026; Weiss et al., 2021). Second, most studies also tested designs with students or general-population participants rather than industrial workers. Third, industrial cobots are most commonly implemented in independent (where the human and cobot are in a shared space but doing different tasks) and simultaneous (where the human and cobot are working on the same workpiece but doing different tasks) scenarios (El Zaatari et al., 2019; Gross & Krenn, 2024). Meanwhile, the most researched scenarios are sequential (human and cobot performing sequential processes on the same workpiece) and supportive (human and cobot work interactively with the same process and the same workpiece) (El Zaatari et al., 2019; Gross & Krenn, 2024). As such, research needs a greater focus on independent and simultaneous scenarios, as that is what industrial workers encounter most frequently. Overall, there is a need to investigate RtH communication design in these scenarios with stakeholder involvement in the design process and validating designs with industrial workers.

1.1 Aims

The aim of this thesis was therefore to design and validate communication solutions for cobot-to-human communication in independent and simultaneous scenarios. The study was twofold, first to generate, refine, and iterate on design solutions using principles from cognitive science and involving relevant industrial stakeholders. Second, to empirically assess the design solutions with industrial workers, with a particular focus on core cognitive ergonomics and user experience outcomes to support the worker.

An overarching research question can be defined as follows:

How can cobot-to-worker communication be designed to support cognitive ergonomics in industrial human-robot collaboration?

This question was investigated by first generating design concepts in workshops with university students and developing the designs with involvement of relevant stakeholders, which addresses the design aspect. Second, the developed designs were assessed in terms of their effects on industrial workers' situation awareness, cognitive workload, perceived safety,

and user experience in comparison to baseline designs, addressing the cognitive ergonomics aspect.

1.2 Outline

This thesis is organised as follows. Section 2 provides a background of necessary information about cobots and their communication. Section 3 presents the theoretical basis of the thesis, including cognitive ergonomics and UX factors. Section 4 summarises previous work on cobot communication solutions. Section 5 outlines the overarching three-phase methodology. This is followed by Sections 6, 7 and 8, which describe the methods and results for each of the three study phases. Section 9 discusses the findings, contributions and implications, methodological limitations, and future research. Finally, Section 10 provides a conclusion for the thesis.

2 Background

This section provides background information on cobots, their industrial scenarios of use and their communication.

A cobot can be defined as a robot that has been designed for human-robot collaboration, and is intended to share workspaces and physically interact with humans (Gross & Krenn, 2024; Hentout et al., 2019). In comparison to traditional industrial robots, cobots do not need perimeter fencing, are lightweight and small, and are flexible in terms of task allocation and location (Hentout et al., 2019). They have a wide variety of possible applications, and the top five most common applications are pick and place, assembly, machine tending, welding, and palletizing (Universal Robots, 2022).

To facilitate safe HRC, cobots also have several safety features. They are power and force limited to increase the safety and minimise the physical risk of the cobot. Safety-monitored stops ensure that the cobot stops if the human enters its workspace, mainly used for situations where the human only occasionally enters the cobot's workspace. For more close interaction, cobots have speed and separation monitoring, enabling the cobot to ensure it does not get closer than allowed protective separation distance but stops if this ever occurs. The separation distance is also based on the speed of the cobot, and closer proximities are allowed at lower speeds. Finally, the human can use hand guiding, usually through a hand-operated device, to communicate commands to the cobot. (Matheson et al., 2019; Shah et al., 2025). The most common appearance of a cobot is in the form of a robotic arm, which can be single or dual. As such, the majority of cobots deployed in industry are non-humanoid and appearance-constrained.

2.1 Collaboration Scenarios

A common classification of human-cobot collaboration in industry is based on the degree of cobot and human intersection in terms of their dependency on each other and their respective tasks. The classifications are (El Zaatari et al., 2019; Keshvarparast et al., 2024):

- Independent: The human and cobot work in parallel, processing two separate tasks on two separate workpieces. The main dependency is maintaining safety between the human and the cobot. This is still a collaborative robot scenario as the cobot and human

are in the same workspace. An example includes a cobot tending to a machine while a human works separately on an assembly task.

- Simultaneous: The human and cobot work on the same workpiece, and at the same time, but carrying out two separate tasks. The main dependency between them is the workpiece, but due to the close proximity, worker physical safety is an important consideration. An example includes a cobot riveting while the human tightens screws on the same workpiece.
- Sequential: The human and cobot work on the same workpiece but carry out two different tasks in a sequential order. The main dependency is the time between the cobot and human tasks. An example is a cobot delivering the workpiece for the human to work on.
- Supportive: The human and cobot work together to process the same task on the same workpiece. The main dependency is the joint task. An example is the cobot holding a workpiece while a human fastens screws.

Most industrial implementations of cobots fall into the first two categories, independent and simultaneous (El Zaatari et al., 2019; Gross & Krenn, 2024). Regarding terminology, in this thesis the human that interacts with the cobot is referred to as either human, worker, or operator. Operator is the traditional jargon and in the context of industrial human-robot collaboration, but can sound misleading, given that it is not always the case that the worker is tending to the cobot as if it were a machine. Instead, with further automation and closer collaboration with more interactive aspects, the cobot and human are a team and several works refer to viewing the cobot as a co-worker (e.g. Bergman et al. (2019) and Gross and Krenn (2024)). Thus, operator is only used when appropriate, and alternatives of human and worker are utilised otherwise.

2.2 Robot-to-Human Communication

Communication within HRI can be considered as the transmission of information between a human and robot (Cha et al., 2018). This communication is bidirectional, but this thesis focuses on the RtH direction of communication. RtH communication includes the robot encoding the information to be transmitted into some sort of signal to be perceived and decoded by the human (Cha et al., 2018). Two important aspects of this signal are content and modality.

2.2.1 Signal Content

There is a lot of potentially relevant information about a robot that can be communicated to human interaction partners. Cha et al. (2018) categorises robot-state information pertinent to co-located humans into the categories of condition, knowledge and activity. The classifications and their subcategories can be seen in Figure 2.1. Condition regards the general working of the robot, including operational (level robot is operating on compared to its optimum functionality), power level, operating mode (between manual and autonomous), physical condition (hardware and its level of functionality), faults (deviations from expected functionality), and affect (emotional state of the robot). Knowledge can be internal, about the robot, or external, about the surrounding world. Internal knowledge includes information about the robot's form, functions and capabilities. External knowledge includes information that is pre-programmed, acquired from sensors, and derived from other available information to the robot. Activity relates to the actions the robot executes (or attempts to), and can relate to either robot availability or task. Availability is whether the robot is able to perform a task. Task information includes goals (desired outcome), plan (how to achieve that outcome), task

progress (development towards the outcome) and success (whether it managed to achieve the outcome or not).

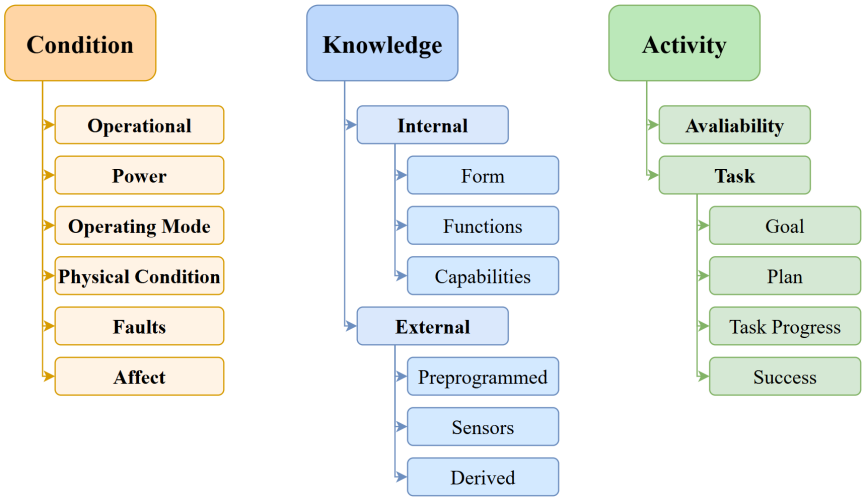


Figure 2.1: Classification of robot-state information relevant to co-located humans, adapted from Cha et al. (2018).

This is not an exhaustive list of information that can be communicated. For example, Cha et al. (2018) does not focus much on reasoning processes behind robot actions, but this will be discussed in Section 3.1.4 on the Situation Awareness-based Agent Transparency (SAT) model (Chen et al., 2014). Still, for RtH communication in industrial cobots, operating mode, physical condition, faults, sensors (e.g. nearby humans and warnings), and all the task-related information are particularly relevant.

2.2.2 Signal Modalities

Signal modalities relevant for the industrial cobot setting include visual, sound, and haptic signals. Given the cognitive and human-centric perspective of this thesis, these are classified based on how the user would perceive them, as opposed to how the cobot would signal them.

Visual

Visual signals include displays such as touchscreens and non-touch screens. These are flexible and can present text, graphics, and images, and function as primary, secondary, mobile, or stationary information sources (Cha et al., 2018; Chen & Barnes, 2021). Prior work has utilised screens (Arntz et al., 2020; Aubert et al., 2018; Hoecherl et al., 2018; Scholz et al., 2025), but interfaces must be well-designed to avoid including too much and fine-grained information.

Light signals are also visual, and can convey quite complex information through variations in colour, light patterns and frequency patterns (Riener et al., 2025). Previous work investigated utilising LED lights on cobots or light modules near the cobot to communicate cobot state information (Arntz et al., 2020; Cao et al., 2024; Dei et al., 2025; Hoecherl et al., 2018; Johnson et al., 2016; Tang et al., 2019).

Cobot motion can also be visually observed. Cobot movement can be overt gestures, or implicit movement trajectories. Overt gestures can be inspired by animation principles like timing and exaggeration (Cha et al., 2018; Takayama et al., 2011), or animal behaviour like signalling task initiation similarly to a dog encouraging their owner to play (Bergman et al., 2019; Sauer et al., 2021). Implicit motion concerns designing movement trajectories to be more legible to the worker, making the movement trajectory easier to predict (Aubert et al., 2018; Dragan et al., 2013).

Sound

Auditory signals include verbal speech and non-verbal sound communication. Industrial noise levels make verbal speech difficult to perceive (Bejerano et al., 2018; Gross & Krenn, 2024; Nielsen et al., 2026). Additionally, previous work has shown that hearing natural speech from a robot can inflate the perceived physical and social capabilities of that robot, which in turn can affect trust and acceptance negatively if those expectations are not met (Cha et al., 2015). Combined, this suggests sounds as opposed to speech may be better suited for industrial environments, and are indeed used today as warnings or alarms (e.g. reversing forklifts) (Cha et al., 2018).

Haptic

Haptic signals are tactile cues that are delivered by wearable means, for example vests, belts or wrist watches (Chen & Barnes, 2021; Villani et al., 2023). They can encode information through frequency, duration and patterns of tactile signals, and previous research has shown that over 12 different cues can be learned in less than 20 minutes, speaking to its effectiveness (Mortimer et al., 2020).

Multimodality

Previous research has shown that utilising multiple modalities can improve performance. For instance, combining auditory signals and visual displays is more effective than only relying on visual displays when completing single tasks (Burke et al., 2006). Combining auditory with haptic, and gesture with sound/light signals can also be beneficial (Cao et al., 2023; Villani et al., 2023).

XR

Finally, extended reality (XR) has, for example, been explored as a way of signalling robot faults (De Pace et al., 2018), a methodology to evaluate cobot signals and test safety (Legler et al., 2023; Mara et al., 2021), and a way to teleoperate robots (Luu et al., 2025). The avenue of XR provides many future opportunities, for instance exploring sensory feedback systems as a way to benefit UX and productivity of HRC (Shourangiz et al., 2025).

3 Theory

This section will discuss the importance of RtH communication for transparency and review some previous cognitive ergonomic cobot research, before moving into the particular cognitive ergonomic issues of SA and workload. The section will then discuss perceived safety and UX, both of which are important for the human-centric view of Industry 5.0 and finally conclude with a short description of the utility of participatory design methodology.

RtH communication is beneficial in HRC contexts as it increases the transparency of robots for the workers. Transparency of different aspects of cobots and robots has been linked to improving several outcomes including acceptance (Concetta Manuela et al., 2025), trust (Cai & Patrick Rau, 2026), team performance (St. Clair & Mataric, 2015), user performance (Ramaraj et al., 2019), mental model accuracy (Ramaraj et al., 2019), perceived safety (Vermeulen et al., 2024), SA (Cai & Patrick Rau, 2026), cognitive workload (Cai & Patrick Rau, 2026; Carissoli et al., 2024), collaboration efficiency (Panagov, 2023), and fluid coordination (Aubert et al., 2018). Thus, cobots having clear and transparent communication to the worker has many potential benefits, including cognitive ergonomic benefits which are of interest for this thesis.

Cognitive ergonomics is a subdomain of the overarching field of ergonomics (also known as human factors) and concerns mental processes in relation to humans and their interaction with other elements of a system, specifically robots when it concerns HRI (Karwowski & Zhang, 2021). Knowledge about the human cognitive factors in such interactions is then used to optimise system performance and worker well-being (Karwowski & Zhang, 2021). In industrial HRC, cognitive ergonomics is less studied in comparison to physical ergonomics (Giallanza et al., 2024; Gualtieri et al., 2021). Still, there are previous works that do consider cognitive ergonomics.

Attahir et al. (2025) and Lorenzini et al. (2023) both review how assessment of cognitive ergonomics can be done in human-robot collaboration in industry. The former also classified ergonomics variables encountered in previous literature into four categories: cobot motion variables (speed, trajectory and jerk), cobot attribute variables (features, payload and inherent sound), collaborative workplace design (proximity, interactions and robot cues), and collaborative task design (task pace, workload, and complexity, cobot as leader/follower, and cobot assisted/not assisted) (Attahir et al., 2025). Each of these were also associated with respective cognitive impacts of vigilance, reaction time, and mental fatigue for cobot motion variables, affective load and perceived safety for cobot attribute variables, SA and anxiety for

collaborative workplace design, and cognitive workload, decision making and frustration for collaborative task design.

Berti et al. (2025) only had a small sample, but as an exploratory study it hints that for low confidence individuals, a presence of a cobot can increase emotional arousal or physiological stress. Cardoso et al. (2021) reviewed ergonomics more broadly but also included cognitive ergonomics and ways it has been measured. The review highlights two main points, that HRC can improve working conditions and that trust is an important aspect of effective collaboration. Concetta Manuela et al. (2025) focuses specifically on acceptance of HRC in industry, and argues that cognitive ergonomic factors are essential for acceptance. The factors mentioned includes cognitive workload, stress, comfort, and perceived safety.

Gualtieri et al. (2022) developed a set of design guidelines for HRC in assembly systems and assessed their effectiveness by adopting and increasing amount of the guidelines and evaluating cognitive factors. Examples of relevant guidelines include to:

- Inform the worker about cobot state and speed
- Avoid velocity variations
- Provide intuitive communication between operator and cobot
- Minimise the amount of feedback interfaces
- Avoid adding human-like features to the cobot.
- Only provide workstation system information when needed
- Demonstrate safety measure's effectiveness and reliability

The full list can be seen in Gualtieri et al. (2022). As progressively more guidelines were adopted in an HRC scenario, Gualtieri et al. (2022) found that trust, usability, perceived enjoyment, and acceptance increased, whereas frustration, stress and cognitive workload decreased. As a note, the guideline on avoiding anthropomorphic features aligns with previous research suggesting increasing a robot's social capabilities risks leading to humans overestimate the physical capabilities of the robot (Cha et al., 2015). Even so, other researchers suggest that humanoid cobots with social cues should be utilised in industrial settings, arguing that overall workplace well-being would be improved (Crivelli & Balconi, 2025).

In the general HRI field, workload and SA are identified as key cognitive ergonomic issues (Bassi et al., 2025; Cai & Patrick Rau, 2026; Chen & Barnes, 2021; Ioniță et al., 2025; Villani et al., 2018). As we saw above, these issues also apply to cobot applications (Attahir et al., 2025; Concetta Manuela et al., 2025; Gualtieri et al., 2022), and will be considered below. Perceived safety and UX are also discussed as they are part of the focus of this thesis and relevant for workplace well-being.

3.1 Situation Awareness

Endsley (1995) described SA as having three ascending levels. Level 1 concerns the perception of elements in the environment, level 2 the comprehension of the elements in terms of the current situation, and level 3 the ability to project future system states. The environmental elements that are relevant depend on the domain and the specific situation (Endsley, 2021). Additionally, the sources of SA vary, for instance coming from different signalling modalities as discussed in the 2.2.2 Signal Modalities section (Endsley, 2021). Importantly, SA in this model is affected by a variety of individual and task factors.

3.1.1 Individual Factors

Individual factors include cognitive mechanisms of information processing such as attention and working memory but also long term memory processes (Endsley, 1995; Endsley, 2021). Industrial environments are complex, and the workers need to select and divide their attention. For visual selective attention, there are four main factors that will attract visual attention from workers: *salience*, *expectancy*, *value* and *effort* (Wickens & Carswell, 2021). *Salient* features are attention-grabbing, by for example being large in size or distinct colours. Attention will also be allocated to areas where it is *expected* to find information. However, not all areas of information are equal, areas can provide information of differing *value* in terms of relevance for tasks and costs of missing information. The final factor of *effort* refers to the difficulty in moving attention, such as glancing towards vs needing to move to a location. To optimise visual attention attraction from the worker, valuable information should be salient, and little effort should be required to see expected and valuable information sources. (Wickens & Carswell, 2021). Still, it is possible to miss critical cues, and additional alarms or warnings (such as distinguishable sound signals (Wogalter et al., 2021)) may be needed to ensure attention capture (Wickens & Carswell, 2021). In more interactive scenarios, such as simultaneous ones, divided attention between the workers task and the cobot may be required. It is easier to divide attention if task demands are low and if the tasks require resources from different perceptual modalities (Wickens & Carswell, 2021). In terms of design, tasks should therefore be simple and not overload a single modality.

The cognitive basis of limitations of task switching and attentional allocation stems in theories about working memory. A prevalent theory is the multi-component model of working memory (Baddeley, 2000; Baddeley & Hitch, 1974). In this model, auditory information is stored in the phonological loop, while visual information is stored in the visuospatial sketchpad. Additionally, auditory and visual noise from the environment can interfere with the phonological loop and visuospatial sketchpad respectively, suggesting an acoustically noisy industry environment could benefit from leveraging visual information. The model also contains a central executive, which manages attention and information integration, and an episodic buffer, which temporarily stores the integrated information. A main characteristic of all components is that they are limited, the central executive in attentional capacity, the episodic buffer in multimodal capacity (Baddeley, 2000), the phonological loop and visuospatial sketchpad in verbal and visual information storage capacity, respectively (Gruszka & Nęcka, 2017). While other models of working memory also exist (e.g. Oberauer (2009)), the limited nature of working memory capacity remains across models (Gruszka & Nęcka, 2017).

For information to be stored in long-term memory, it need to be rehearsed over time (Atkinson & Shiffrin, 1968), meaning more experienced operators have cognitively established schemas and mental models that can support their SA in a particular situation (Endsley, 1995; Endsley, 2021). Other individual factors that affect SA in a particular situation includes goals, objectives, and preconceptions that relate to the situation (Endsley, 2021). These are especially relevant in the second level of SA, because goals and expectations will affect the understanding of perceived environmental elements. Environmental elements will be given significance and interpreted based on the worker's active goals.

Overall, in relation to SA, selective/divided attention and means of capturing it is particularly important in the first level of SA, for the worker to perceive the most relevant elements of the environment. In the cobot collaboration setting, the elements include the RtH communication and various artifacts (materials, visible faults, machines, tools etc). The limitations of working memory are relevant in both the first and second levels. In the first level it is important to ensure incoming information does not exceed the working memory capacity. In the second level working memory is also relevant as perceived elements need to be placed in context and integrated with long-term memories. In cobot contexts the second SA level could be understanding the elements of red signal and empty storage container as a missing materials error for instance. Goals, expectations, and expertise is particularly important in

the second and third levels of SA. For the second level, as previously mentioned, goals affect how the situation is interpreted, and expertise means relevant mental models can be leveraged. Further, for the third level of SA, established mental models and training will aid in predicting what will happen in the system and what potential consequences exist, such as consequences of refilling the cobot's material now or after the active stage of the worker's task.

3.1.2 Task Factors

Task-related factors include system capability, interface design, stress and cognitive workload, complexity, and automation. System capabilities refer to the capabilities for the system to gather and present needed information, and interface design concerns the way this information is presented. Physical (e.g. heat, fatigue) and psychological (e.g. fear, uncertainty) stressors can negatively affect SA, and high cognitive workload has implications for SA. SA and workload interact in the ways seen in Table 3.1. Complexity refers to the complexity of the system itself, with more elements and dynamic relationships between them, the operator's task may also become more complex. How much this affects the workload can depend on the expertise of the worker, but it is possible their SA suffers. Automation can eliminate excessive workload for the operator; however, the operator will also be out-of-the-loop and thus have lower SA than otherwise. This is mainly a problem when the autonomous system fails and the human needs to intervene without sufficient SA. (Endsley, 1995; Endsley, 2021). Given the autonomous nature of cobots, this is a concern in the current thesis and further highlights the importance of providing transparency information for the worker to counteract the out-of-loop effect.

		Situation Awareness	
		Low	**High**
Workload	**Low**	Operator has poor idea of the situation, but also not trying to improve the awareness (e.g. due to inattentiveness or low motivation)	Ideal state - operator easily acquires the required information
	High	Operator can only attend to some information or could be failing to achieve SA due to perception/integration problems as a consequence of the high workload.	The operator works hard, but is still achieving a complete picture of the situation

Figure 3.1: Relationship between situation awareness and workload, adapted from Endsley (1995).

3.1.3 Theory Justification

A critique of Endsley's (1995) model of SA is that it does not support understanding of the distributed and dynamic aspect of socio-technical systems (Stanton et al., 2010). Endsley herself rejects the notion that the model cannot explain dynamic systems, stating that the feedback mechanisms are integrated in the model. As such, observing the environment to

update the model of the situation allows dynamically updating search goals for information (Endsley, 2015). Still, Endsley (2015) does reject the idea that inanimate objects have "awareness", which may well be a limitation in intelligent and artificial intelligence (AI) systems. Indeed, as mentioned previously, a lot of focus in the cobot field has been put on the cobot having a better model of the world around it, including being "aware" of human signals (El Zaatari et al., 2019). As such, this limitation is valid, and other models of SA may better account for situated and distributed facets. Situated situation awareness posits that operators heavily rely on tools in the environment to maintain SA, thus the focus lies on the human and their interaction with the task environment (Chiappe et al., 2012). Other models focus on team or shared SA, defined by whether team members have the required and same SA information respectively (Endsley, 2015). Distributed situation awareness (DSA) posits that SA is not held by any individual operator, but rather SA is distributed across the system (Salmon et al., 2008). Models like these have notable strengths in light of cognition theories of 4E and distributed cognition theories, and may be valuable perspectives in complex and dynamic systems (Clark, 2014; Clark & Chalmers, 1998; Foglia & Wilson, 2013; Hutchins & Klausen, 1996; Varela et al., 1992).

This thesis does not reject these general theories of cognition, but nonetheless mainly utilises Endsley's (1995) model of SA. This is for two main reasons. First the focus of this thesis is on RtH communication and on scenarios including one worker. Thus, the current thesis is concerned with the *human* gaining information about the robot operation, as such, the inclusion of the awareness of the system as a whole, or other agents within it (as in DSA and team/shared SA) is not necessary. When it comes to situated SA, the core idea aligns with the current focus in the sense that having clear cobot signals can be seen as a tool for operator to offload mental processing required to acquire SA. While this can be seen as an additional benefit of intuitive and well-designed interfaces, it is not sufficient cause to be used as a main theoretical framework in the cobot scenarios for this thesis. Secondly, Endsley's (1995) model has strong ties to a related and relevant theory of agent transparency which describes transparency information in interface design, and is presented in the following section.

3.1.4 SAT model

Endsley's (1995) model of SA demonstrates the importance gathering information for SA. Another theory, Situation Awareness–Based Agent Transparency (SAT), bases itself on Endsley's (1995) levels of SA, but instead describes transparency requirements for each (Chen et al., 2014). The authors define agent transparency as "the descriptive quality of an interface pertaining to its abilities to afford an operator's comprehension about an intelligent agent's intent, performance, future plans, and reasoning process" (Chen et al., 2014, p. 2). To support the operators at each level of SA the operator should be provided with:

- Level 1 – "basic information about the agent's current state and goals, intentions and proposed actions" (Chen et al., 2014, p. 2)
- Level 2 – "information about the agent's reasoning process behind those actions and the constraints/affordances that the agent considers when planning those actions" (Chen et al., 2014, pp. 2–3)
- Level 3 – "information regarding the agent's projection of the future state, such as predicted consequences, likelihood of success/failure, and any uncertainty associated with the aforementioned projections" (Chen et al., 2014, p. 3)

Incorporating these levels into interface design can provide good transparency and aid the operator's understanding of agent reasoning as well as decision-making about potential operator intervention (Chen et al., 2014). Though, the authors point out that all levels may not be needed to achieve transparency, but they are different aspects of it. The examples and applications described in the original publication of this theory surround military operation and

map-based human-agent teaming (Chen et al., 2014). Still, it provides a transparency framework and has at least one clearly relevant level since level 1 describes similar information as the 2.2.1 Signal Content section above. Additionally, later expansions of the theory add autonomous robots explicitly as a target agent for the theory, but focus on mixed-initiative systems where the human and agent collaborate in decision-making (Chen et al., 2018).

3.2 Cognitive Workload

A concern with transparency is the risk of providing too much information to the operator, increasing their cognitive workload (also known as mental workload (Hancock et al., 2021)). Cognitive workload pertains to an operator's allocation of limited internal processing resources or capacity to meet external task demands (Hancock et al., 2021). A prominent theory of cognitive workload is multiple resource theory (Hancock et al., 2021; Wickens, 2008). It posits that there are metaphorical reserves of energy to be used for cognitive processing, and different cognitive tasks draw from different pools or dimensions. The dimensions are: stages of processing (perception, cognition (which is limited as described above), or responding), codes of processing (spatial vs verbal), and perceptual modalities (audio vs visual). If two tasks utilise the same aspect of any given dimension, for example engaging in two spatial cognition tasks, the tasks will be more difficult to perform, and the pool of resources will be drained faster. When resources are drained the operator reaches cognitive overload, and task performance suffers. However, if the tasks are different on one or more of these dimensions they draw from different resources which allows parallel processing (more efficient multitasking). Still, overall, the resources are limited and the higher the task demands (e.g. due to complexity or novelty of the task) the faster the resources drain. (Hancock et al., 2021; Wickens, 2008).

Given the limited resources available for cognitive processing, reducing the risk of cognitive overload is important. This is pertinent in industrial HRI as previous studies have shown that working with a robot can increase cognitive workload (Arai et al., 2010; Bassi et al., 2025; Carissoli et al., 2024). Just as clear and transparent RtH communication can support SA, it can aid in reducing cognitive workload as well, further emphasising the requirement for good interface design (Carissoli et al., 2024; Chen et al., 2014; Gualtieri et al., 2022; Keyes et al., 2010; Villani et al., 2018). For example, top-down elements that build affordances into the interface allows for intuitive understanding, requiring less resources (Chen et al., 2014). Gualtieri et al. (2022) implemented several guidelines for cognitive ergonomics in cobots and showed that predictability and familiarity also have positive effects on cognitive workload. Finally, the theories of workload and working memory themselves suggest ways of mitigating workload, such as ensuring parallel tasks are sufficiently different in terms of cognitive resources, and that working memory limitations should be considered.

3.3 Perceived Safety

Safety is a key concern within ergonomics as well as in industrial environments. Physical safety of the worker who is collaborating with the cobot is always important and takes a prominent role in more interactive scenarios (simultaneous, supportive and sequential) (Keshvarparast et al., 2024). However, the perceived safety of a situation is also important for the worker, as not perceiving a situation as safe would contribute to a negative experience. Previous work has shown that workers' psychological safety relies on the cobot being able to clearly communicate what the cobot will do next (Novakazi et al., 2025; Vermeulen et al., 2024)[1]. Indeed, another study also found positive effects of transparency on perceived safety (Kluy & Roesler, 2021). In HRI perceived safety is affected by several factors including context of use (such as robot role and task), user comfort (e.g. robot moving too fast or too

[1]Novakazi et al. (2025) is currently under review, obtained and used here with author permission

close), experience and familiarity (creates more positive attitudes), predictability and transparency (as aforementioned), sense of control (being in charge of the robotic system), and trust (many robot factors, e.g. reliability and level of autonomy affects trust) (Akalin, 2022). Overall, clear communication can contribute to better perceived safety of the cobot, which is an important factor in the user's experience.

3.4 User Experience and Participatory Design

Given the human-centric nature of Industry 5.0 as discussed in the introduction, another aspect to consider in human-cobot collaboration is UX. UX can be understood in terms of three facets: beyond the instrumental, emotion and affect, and the experiential (Hassenzahl & Tractinsky, 2006). The beyond the instrumental facet means that UX is something beyond the practical value of the tool as holistic, aesthetic, and hedonic aspects of designing products also need to be considered. Emotion and affect highlights considerations such as subjective well-being and positive emotions like joy. Further the affective quality of a product can be an antecedent to other usability outcomes (e.g. usefulness and intention to use). Additionally, emotion can be a consequence of interacting with a product, but there are difficulties in designing products with a specific emotional outcome in mind. The experiential facet regards the nature of an experience, of it being dynamic, complex, situated, unique, and temporary. Still, UX is also influenced by usability and functionality, which should not be overlooked. (Hassenzahl & Tractinsky, 2006).

Overall, UX aims to improve the humans experience of the product, which includes both the practical use of a product and the hedonic value of using it. A related aspect to UX in HRI is expectations, and it has been suggested that if expectations of robots are not met this can lead to a worse UX (Rosén, 2021). Also in relation to UX, Rosén (2021) notes that considering UX too late in the design process, or even as an afterthought to a designed product, is of little use. Therefore, to achieve a good UX, it needs to be considered before high-fidelity prototypes or finished products.

Another aspect highlighted as important for success of a design solution is stakeholder involvement in the design process, which can be facilitated through participatory design methods (Schuler & Namioka, 1993). Participatory design started on the sentiment that people affected by a decision should get a say, just as users are affected by the design of a product and therefore should get a say in the design process (Schuler & Namioka, 1993). This means involving stakeholders (end users, management, and other impacted individuals) in the design process, for example through co-design, where designers and stakeholders work together to produce the design (Hornbæk et al., 2025). Several authors argue for the importance and value of participatory design within human-robot collaboration in industry (Cao et al., 2024; Nielsen et al., 2026; Scholz et al., 2025; Weiss et al., 2021).

4 Related Work

Several previous works specifically investigate design solutions for cobot-to-human communication in industry. These are summarised in Table 4.1, but the most relevant for the current thesis are described in more detail below.

Scholz et al. (2025) is the previous work most closely aligned with the aims of this thesis. A flexible OLED screen that was mounted around the cobot arm was proposed as a RtH communication solution. They conducted co-design workshops with workers to determine how and what information should be communicated and implemented on the screen. In order of priority, this information was: robot movement, process monitoring, working steps, robot error, information on parts, stock information, ergonomic feedback, and general information. As for how information should be displayed, the overall preference was for colours and simple illustrations. These suggestions were implemented and a larger study with novel workers who saw cobot interactions (with or without the OLED screen) on video was conducted. The results showed that the OLED screen was significantly better than the control in terms of anxiety, attitude towards the robot, perceived enjoyment, intention to use, trust, and social influence, but did not differ from the control on perceived usefulness or sociability.

Cao et al. (2024) also conducted a co-design study with industrial employees but utilised a light-based solution for RtH communication, with a light module mounted on the cobot. The light module was LED lights inside a round light diffuser, allowing for 180-degree visibility. The light module was used to communicate the cobot's state (moving, rotating, and reaching destination), ergonomic feedback to the operator, and the amount of force being applied to the workpiece. The light module was part of a larger workstation design and was therefore not quantitatively validated on its own. Still, feedback from workers included that the light module was useful in understanding the cobot but they also suggested implementation improvements. This feedback was based on semi-structured interviews with four employees who were part of the co-creation process.

Table 4.1: Overview of previous research on cobot-to-human communication design solutions.

Reference	Scenario	Participants	Signalling	What was communicated	Quantitative outcome measures	Interesting results or notes
Scholz et al. (2025)	Supportive	Operators	OLED screen around cobot arm	Cobot movement, process, working steps and assembly instructions, information on parts, error, stock information, ergonomic feedback	Preference, anxiety, attitude, perceived usefulness, intention to use, perceived enjoyment, perceived sociability, social influence, and trust	OLED improved anxiety, attitude towards the robot, perceived enjoyment, intention to use, trust, and social influence, but did not differ in perceived usefulness or sociability compared to control without OLED screen. This was a co-design study.
Tang et al. (2019)	Independent	University volunteers	LED strip around cobot arm vs tower light	Cobot state	Reaction time, hit-rate, task performance, ease of monitoring, level of tiredness, visual fixation and dwell	Having LED lights directly on the cobot as opposed to traditional tower lights improved reaction times, hit rates, and ease of monitoring, while reducing eyesight diversion from the operator's task.
Arntz et al. (2020)	Sequential	Majority university students, some got course credit	Light, screen, and gestures	Displays: cobot action explanations, working steps, collision warning, movement, production output and remaining time. Lights and gestures: cobot state	Perceived stress, frustration (insecure, discouraged, irritated, stressed, annoyed), and social presence	Participants had lower perceived stress, more positive emotions, and perceived the cobot as more socially present in the condition with the screen, lights and gestures, as opposed to without. Experiment in VR.
Cao et al. (2024)	Supportive	Industrial employees	Lights from a round light module mounted on cobot	Initiation, task specific information, finish, force feedback (that the worker is using on the component), and bad worker posture	HRC quality (but perceived usefulness, perceived ease of use and intention to use were assessed qualitatively)	The light module was part of a larger workstation design and was not validated on its own. Still, the light module received both positive feedback and suggestions from workers.
Dei et al. (2025)	Independent and supportive	General participants (and participants with autism spectrum disorder)	Lights and sound from a free-standing module	Cobot state (error, searching for part, found piece, continued workflow)	Regarding the light module: perceived function, preference of working with or without it, whether it was disturbing, if information was intuitive, and where the device should be positioned	Signals were understood with varying success (error and found piece were easily identified but other states were more difficult). Also only half the participants definitively preferred the device and understood its function.
Cao et al. (2023)	Sequential	Online recruits	Light module near effector, sound, and gestures	Feedback to worker: cobot in neutral state but active, worker doing correct actions, worker doing something incorrect, cobot needs to hand over component to worker	Godspeed (Bartneck et al., 2009): anthropomorphism, animacy, likability, perceived intelligence, perceived safety. Also: perceived enjoyment, perceived usefulness, intention to use, and recognition rate	Compared to only gestures, utilising all signalling modes had higher scores for anthropomorphism, animacy, likability, perceived intelligence, perceived enjoyment, and intention to use, but no difference for perceived safety or perceived usefulness. Only utilising head gestures had a significantly lower recognition rate compared to multimodal, gesture-and-light, and gesture-and-sound signals.
Aubert et al. (2018)	Supportive or simultaneous	General participants	Legible movement and screen	Where the cobot will reach towards	Task measures (participant idle time and number of conflicts), subjective team fluency, and subjective teammate transparency	Both intent communication types supported HRC compared to the control, but the most effective strategy was to have both movement- and display-based intent communication.
Hoecherl et al. (2018)	Sequential	Mostly students, some participants indicated production experience	Visual (lights and screens) and auditory (messages and sounds)	Cobot: status, errors, goal position, progress, mode of operation and limits	User experience, preferred communication channel, signal understandability, successful perception of progress bars/process steps/multiple warnings, preference for more extensive warnings, sound relevance, ability to differentiate sound, and general feedback	Compared to working with a human, the user experience was better when working with a cobot, but efficiency was perceived as lower. Most participants preferred the primary screen and understood signals. Some signals were missed, but the progress bar on the primary screen was consistently perceived. Sounds were identified as relevant.
Sauer et al. (2021)	Sequential	General participants (convenience sample)	Gestures (zoomorphic and abstract) and lights on cobot base	Greeting the user, prompting the user to take part, waiting for a new command, error: storage container is empty, and shutdown	Godspeed (Bartneck et al., 2009), correctly identified signals, preference, user experience (attractiveness, joy of use, intuitive use, intention to use)	Zoomorphic gestures scored higher than alternatives in anthropomorphism, animacy, likeability, attractiveness, joy of use, and intuitive use, but lower in perceived safety. Both gesture-based signals outperformed light displays in perceived intelligence and were classified more accurately, while zoomorphic gestures were only better than light displays on intention to use.
Villani et al. (2023)	Independent and supportive	Students and researchers	Haptic wrist band and lights	Laboratory experiment: haptic information for which part the human should pick and bring to the cobot. VR experiment: lights and/or haptics signal changes in task allocation between cobot and human	Laboratory experiment: assembly time. VR experiment: usability and reliance on haptics/lights when provided together	Assembly time improved with haptic feedback compared to without. The VR experiment showed haptic signals had better notification detection than lights, but providing both was best. In terms of usability, vibration had higher scores, but when both modalities were available the majority of participants relied on both.

Aubert et al. (2018) investigated the effect of intent communication modalities on team fluidity. The modalities investigated were movement, through predictable and legible movement design, and a screen-based display. The robot's task was not described, and as such the task can be classified as either simultaneous or supportive depending on if the robot has a different or the same task as the human, respectively. The results showed that both types of intent communication support HRC compared to the control, but the most effective strategy was to have both movement and display based intent communication.

Other cobot signalling solutions include using LED lights around the cobot arm for state indication in an independent scenario (Johnson et al., 2016; Tang et al., 2019). Having LED lights directly on the cobot as opposed to traditional tower lights improved reaction times, hit rates, and ease of monitoring, while reducing eyesight diversion from the operator's task. Qualitative results included that most participants preferred the LED solution because it was more attention-grabbing.

As can be seen in Table 4.1, independent and simultaneous scenarios are underrepresented. This is problematic because independent and simultaneous scenarios are the most common industrial implementations (El Zaatari et al., 2019; Gross & Krenn, 2024). This is reflected in Gross and Krenn (2024), where the design guidelines given for these scenarios are sparse and several are based on research with robots outside the industrial context (e.g. service robots (Koay et al., 2007, 2014)), or even with humans (proxemics recommendations based on Hall (1966)).

Further, Table 4.1 shows that most studies do not utilise participatory methods or test designs with industrial employees. Several previous and relatively recent works point out the need and value of involving stakeholders in the design process (Cao et al., 2024; Nielsen et al., 2026; Weiss et al., 2021). Participatory design strategies are important for hearing the voices of operators, shift leaders, production technicians and so forth (Weiss et al., 2021). While industrial workers are difficult to reach as a participant group (Nielsen et al., 2026), it is important to test and evaluate design solutions with them. They bring experience from industrial factories meaning they have a better opportunity to provide context-relevant opinions and suggestions. They may also be aware of general attitudes towards robots or certain solutions among industrial workers. Additionally, they would have better insights into cognitive factors, such as those of interest in this thesis. This is because they are aware of the distractions, common work stresses, and safety concerns that a generic participant, such as a university student, would not necessarily be aware of.

Overall, there is a need for participatory design of cobot-to-human communication in independent and simultaneous scenarios, and testing designs on industrial workers. The methodology for this thesis was designed to achieve this and the methodological approach is discussed in the next section.

5 Method Overview

The study utilises research through design (RtD), in which the aim is to develop theoretical knowledge through design activities (Anderson & Shattuck, 2012; Nelson, 2013). This is particularly well-aligned with the current aims as the study seeks to advance knowledge about cobot signalling. Further, the core principles of RtD include a participatory and iterative design process, which allows evolution and development of the design through involvement of relevant stakeholders from the context in which the design is to be used (Anderson & Shattuck, 2012; Nelson, 2013). This is also well suited for this thesis as involvement of stakeholders and workers from industrial contexts is a central aim. The RtD methodology is therefore appropriate for both the design development and empirical assessment objectives of the current thesis.

In keeping with RtD principles, this study employed a three-phase study design consisting of concept generation, expert evaluation, and empirical assessment of the developed concepts (Figure 5.1). The first two phases were focused on developing two designs for cobot communication, one for independent scenarios and one for simultaneous scenarios. The final phase aimed to empirically validate the developed designs. The concept generation involved holding design workshops with university students and subsequently refining the designs into a cohesive design language (sketch aesthetics, fonts, and so on) and distinct concepts. The second phase involved presenting the refined concepts to be evaluated by industry professionals and then narrowing down and developing concepts based on their feedback. The final phase involved assessing two developed concepts, one for an independent scenario and one for a simultaneous scenario. This was done through a mainly quantitative questionnaire administered to processing and assembly workers, which assessed several cognitive ergonomics factors as well as UX. As per RtD, the industrial context was considered in each phase, and relevant stakeholders were involved in the second and third phases, contributing to the participatory design process. As for the iterative principle of RtD, these three phases were sequential and between each phase the concepts evolved and developed based on insights from the previous stage. The methodology and outcomes of each phase are detailed in their respective sections below.

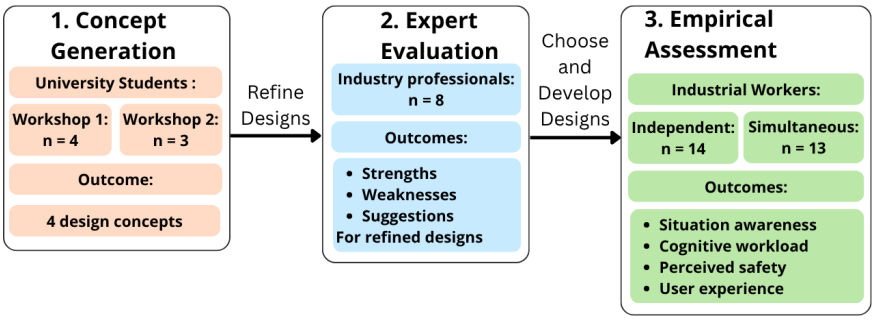


Figure 5.1: Methodology overview showing the chronological phases, sample sizes, and main outcomes for each phase.

5.1 Ethics

Informed consent was gathered from all participants included in the study. Two participants who filled out the empirical assessment questionnaire had to be excluded as they did not mark the consent question on the first page. The consent information provided for each phase of the study can be seen in the appendices. Informed consent should always be taken very seriously and was of particular concern in the current study as distribution of the empirical questionnaire to workers in the majority of cases was done by a manager or boss. This may lead to the worker feeling they must follow a directive since they are on paid time. Whenever the researcher was present, it was always ensured the participant read the consent information, and when recruiting participants on the shop floor it was always verbally ensured the participants volunteered to fill out the questionnaire. Demographic data was only collected when deemed necessary, which was only for the empirical assessment to assess diversity of sample and not the other phases of the study. Additionally, company contacts were offered either the final report or a summary of the results as follow-up to the study participation, which all contacts appreciated and wished to receive.

6 Concept Generation

Concept generation was achieved through two workshops conducted with university students. The aim was to generate several distinct design concepts for cobot signalling in industrial scenarios. The workshops took approximately two hours each to complete.

6.1 Method

The workshops were based on the participatory method of future workshops. The method is used to involve stakeholders in addressing faced challenges through the ideation of a future situation and the steps taken to realise that future (Hornbæk et al., 2025; Raider-Roth et al., 2021). The method involves three phases: critique, fantasy (also known as utopian), and implementation (also known as realisation). The critique phase involves critiquing the current situation. The fantasy phase involves addressing the important critiques by creating a conceptualisation of the ideal future. In the implementation phase there is a focus on concrete steps that need to be taken towards realising the ideal future. (Hornbæk et al., 2025; Raider-Roth et al., 2021). The method was slightly adapted for the second workshop, which is described in the 6.1.3 Procedure section below.

This workshop method was chosen over other more expressive workshop methods like acting situations of use mainly due to ease of data collection (drawings vs acted out situations) and to improve engagement through a lower barrier to express ideas (Hornbæk et al., 2025). In workshop situations with a small number of participants, of whom you may not know well, acting out and role-playing scenarios may not be desirable for participants. Additionally, role-playing may be more suited for service or less appearance constrained products. Representing cobot signalling through human movement and role-playing may inadvertently exclude some types of signalling that humans are not inherently capable of. For instance, light signals may not come to mind due to being more difficult to represent than speech or movement.

6.1.1 Participants

Workshop 1 had four participants, all enrolled in a course in human-computer interaction at a Swedish university. Workshop 2 had three participants, all from a master's program in cognitive science at a different Swedish university. No demographic information was collected,

both to protect participant anonymity and because the aims were not to compare or analyse group effects. Participants were recruited through convenience sampling.

While it would have been desirable to also include relevant stakeholders in the workshops, the logistical constraints meant this was not feasible. Including workers is particularly difficult given the constraints posed by the factory operation which prevents them from participating during working hours. Given that the context of the thesis did not make it possible to provide compensation, recruiting workers for their spare time was also not realistic. Including managerial staff was also impossible given the constraints of potential participants and the time frame of the study. Students with backgrounds in human-computer interaction and cognitive science were therefore considered a reasonable alternative. Additionally, the entirety of this study is nested within the early stages of a design process, within the concept phase (Arvola, 2020), before any higher fidelity prototyping or major feasibility considerations. This means that, even if stakeholders are not involved in the first phase of this study, they are still being involved early on in the overall design process.

6.1.2 Materials

A template in A3 paper format was provided for the participants to document their final designs. This final design template contained a central area for drawing the design, and four surrounding boxes, each with a question. The purpose of the questions was to encourage the participants to reflect and explain their thinking behind the designs. The questions were: "What were the most important decisions made? How were they made", "How does the design solve the presented problem", "What were the hardest aspects to solve in the design problem", and "What are the limitations of your design". The template can be seen in Appendix 11.1.

The participants were also provided with coloured pencils, coloured markers, ink pens, plain A4 paper, and a few copies of the A3 design template to sketch their ideas.

All participants gave their consent to participate, and the full form can be found in Appendix 11.2.

6.1.3 Procedure

The procedures for the two workshops varied slightly but both common and workshop-specific procedures are described below.

Common Procedure

The introduction and end of the workshops were the same for both sessions. The workshops started by the facilitator presenting the agenda. The facilitator then presented background information on cobots within industry, the perspective and task for the workshop, and the industrial context to be considered.

For the background, first, a definition of cobots was presented (Hentout et al., 2019, pp. 766–767) along with a historical overview of cobots (Shah et al., 2025) and some examples of their application areas (Patil et al., 2023). The background also contained information that cobots have adaptable implementation and could perform varying tasks at an industrial workplace. This was illustrated with stills from Productive Robotics demonstration videos of welding, machine tending, and parts manipulation from their YouTube channel.[1] It was also described that cobots are capable of sharing spaces with, understanding, and engaging

[1] Welding application: https://www.youtube.com/watch?v=5tOSAmMZQvU&t=58s (Productive Robotics, 2024).
Machine tending application: https://www.youtube.com/watch?v=HrTFCaQdjqU&t=289s (Productive Robotics, 2023).
Parts manipulation application: https://www.youtube.com/watch?v=SEqnzGDf3f4&t=66s (Productive Robotics, 2025).

in fluid collaboration with humans. This included stating that it is possible to view the cobot as a co-worker and the importance of supporting worker UX. It was further emphasised that part of a good UX is ensuring the human can understand the cobot's behaviour, such as its movement, goals, progress, or states (Cha et al., 2018). The overarching workshop task was presented through posing the main question: "How can we physically design the cobot to better communicate relevant behaviour to the worker?" Participants also viewed a short video showing an OB7 cobot loading and unloading a computer numerical control (CNC) machine to, in the first workshop, see a specific scenario to consider and, in the second workshop, gain an idea of how a cobot moves[2]. Finally, the industrial context was illustrated with a YouTube video showing a Shantui bulldozer production process with high noise levels and moving parts.[3]

At the end of both workshops there was an opportunity for the participants to discuss and share their thoughts about the workshop. A presentation slide prompted discussion with the words: "Concerns?", "Thoughts?", and "Difficulties?".

Procedure Specific to Workshop 1

Following the introduction, the main workshop task started. For the first workshop this followed the future workshop format, thus the first step was the critique phase. Participants were instructed to identify and write down negative aspects about the current situation (as presented in previous videos). They were also instructed to consider how the future would look if these things were not amended, and finally to mark the most important of their critiques.

After the critique phase, participants received instructions for the fantasy phase. First, participants were tasked to individually come up with ideas on how the cobot could communicate while avoiding issues identified in previous step. Instructions also included that they should think creatively about an ideal scenario and draw or describe their ideas. Second, participants were tasked with presenting their ideas to the group and discuss and decide on their favourite ideas.

After the fantasy phase, participants received instructions for the implementation phase. They were tasked to develop their chosen concept/s while considering what would be realistic to implement. They were also instructed to illustrate their ideas so the group can more easily visualise the concept. The slide reminded participants to think of how the concept would work in a human-cobot interaction scenario. Finally, they were instructed to fill out the A3 paper for their final design. A presentation slide with some guiding questions in case participants got stuck in their discussion was also presented and kept up on the projector for the latter half of the phase.

Issues Encountered During Workshop 1

It was clear after the first workshop that the noisy and busy context was not sufficiently emphasised in the slides as this was not something participants discussed or considered in their final design concept. Additionally, the slides did not provide enough reminders to keep specifically to designing the cobot signalling, as discovered through participants including other aspects such as emergency stop buttons in their design. Thus, for the second workshop, to better target the intended design problem, the format was slightly changed.

Procedure Specific to Workshop 2

Firstly, a context critique phase was integrated in the presentation so participants could engage with specific challenges faced in the context. The instructions were to identify aspects

[2]Cobot movement: https://www.youtube.com/watch?v=HrTFCaQdjqU&t=219s (Productive Robotics, 2023)

[3]Factory context from Shantui Construction Machinery Co., Ltd. (2026): https://www.youtube.com/watch?v=NC7vEpnW0xo&t=1036s (Satisfying Tech, 2025).

that make the human-robot communication difficult in the context video [3]. Specifically, two questions were posed to the participants: "What can interfere with communication?" and "What types of or ways of communication would be the most difficult in this context? Why?". The slide also instructed participants to write down as many issues as possible and then mark the most important ones.

Secondly, participants were presented with a specific independent and simultaneous scenario allowing for a clearer and more focused task. Independent and simultaneous scenarios were chosen as this was the focus of this thesis. The instructions for the participants still retained elements from the fantasy and implementation stages. Specifically, participants were instructed to first individually write down what the cobot should communicate and creatively brainstorm how the cobot should communicate that information (fantasy phase-adjacent and same questions as posed by Scholz et al. (2025)). Secondly, the slide instructed participants to work together, first by presenting their ideas and picking the most promising. Here they were also tasked to consider feasibility in the context, how the interaction could look and when the communication should happen (implementation phase-adjacent). Finally, the participants were instructed to draw the concepts on the A3 final design template.

The independent scenario was presented first [4] and after the design concepts were completed, the simultaneous scenario was presented [5] and the same procedure followed.

6.2 Outcomes

The outcome of the workshops was four distinct design concepts that can be seen in Figure 6.1.

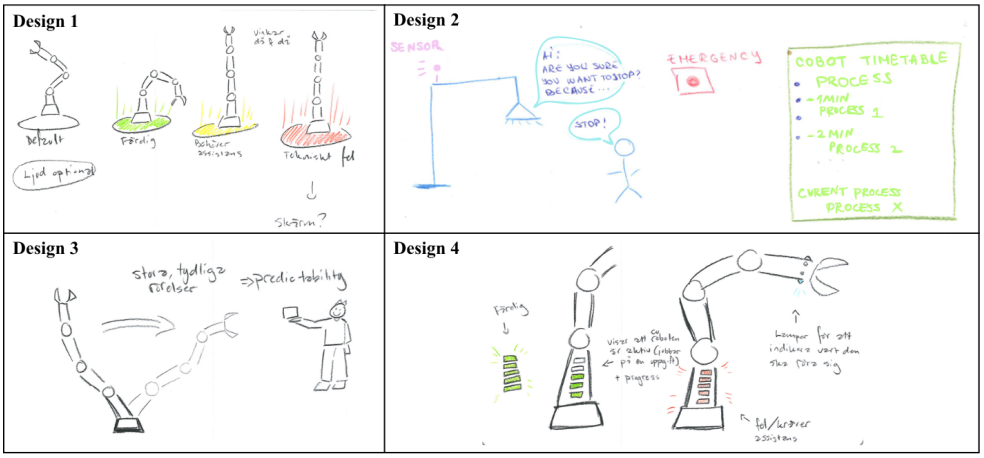


Figure 6.1: Raw designs from workshops. Designs 1 and 2 are for independent scenarios, and 3 and 4 for simultaneous.

Design 1 (Figure 6.1) was for the independent scenario from Workshop 2. The participants decided to design light signals from the cobot base and gestures to attract attention to the cobot and signal important states. The rationale included to use the visual modality, since the industrial context is noisy, and that the signals should be clear, large, and easy to see in one's peripheral vision, since the worker divides their attention between monitoring the cobot and their own task. The participants developed four distinct signalling states. First a default state, which is when the cobot is working normally. In the default state there would be no signalling from the cobot, the rationale being to not overload the worker with unnecessary information.

[4] Timestamps 0:14-0:43 was shown for overall context and 2:06-2:20 for the independent scenario: https://www.universal-robots.com/case-stories/sensata-technologies/ (Universal Robots, 2025)

[5] Timestamps 1:49-2:06: https://www.youtube.com/watch?v=uqdrlXuIEFM (FANUC America Corporation, 2020)

Second, a finished state, when the cobot has finished its current task successfully. To signal that it is finished, it would slump down and the base would light up in green. Third, a need assistance state, when the cobot needs help from a human to proceed with its task. The cobot would signal this through extending straight up and the base would light up yellow. Fourth, a technical error state, when there is a more severe issue with the cobot such as it being damaged or having a malfunctioning part. The cobot would extend straight up as in the previous state, but the base would light up red instead of yellow. For both the third and fourth states, the cobot would start waving after a while to further attract attention if it has not already been checked on. Additionally, there would be an accompanying screen that details what the error or need for assistance is. Participants also discussed the utility of additional sound signals, but did not end up integrating it in the design besides saying it is an optional feature. If such a feature was implemented it would need to be a distinctive and at the correct pitch for perceiving it even with the noisy environment and potential hearing protection. The limitations of this design that the participants identified were that it would not work well for colour-blind individuals, and that the waving gesture may be interpreted differently in different cultures and thus may not be universally understood.

Design 2 (Figure 6.1) was from Workshop 1, where an independent scenario was shown. The communication aspects of the design included spoken language, and a separate screen with a cobot timetable. The cobot would both produce and understand speech. It would speak to the worker to tell them about its states or to answer a verbal command or inquiry from the worker. Participants suggested the verbal communication would be implemented through the use of AI. A timetable of the cobots current and upcoming processes, and their related timings, would be displayed on a screen separate from but near the cobot. Outside of the communication focus the participants discussed safety and included both cobot sensors and an emergency stop button in their design. The limitations the participants identified included that the cobot is "still a machine" and that there can be errors because human behaviour is heuristic and unexpected.

Design 3 (Figure 6.1) was for the simultaneous scenario from Workshop 2. For this design the participants identified that the cobot's upcoming movement should be obvious to the worker, for their comfort and understanding of the cobot. They focused on discussing that the cobots intended movement should be predictable to the human working with it. The concept idea was that the cobot should use large sweeping motions as opposed to the most optimal route, and to lead the movement with the effector to indicate the direction of movement. The participants identified that a limitation can be that the movement becomes less energy efficient.

Design 4 (Figure 6.1) was also for the simultaneous scenario from Workshop 2. As opposed to the independent scenario, they reasoned that, due to the closer proximity, the signalling did not need to be as large as in the independent scenario. Still, movement intent should be clear to the worker just as in Design 3. They also wanted to ensure it is easy to tell when the cobot is active and not, such that the workers know when they need to be vigilant and not. Further, indicating progress in addition to the states of error and completion (as in design 1) was identified as an important piece of information. These considerations led to the integrated indicator. The indicator has several levels, and filling of the levels indicates task progress. The progress is indicated in green, which also signals that the cobot is active. When the cobot has finished its task, the entire indicator is filled. If the cobot encounters any error, the indicator instead turns into a solid red, which signals that it needs assistance. It would also double as an indicator that the cobot is not moving and thus indicating to the worker that it does not need to watch out for upcoming movement. The participants also included movement indication on the design through a band of lights around the outermost part of the cobot arm. The lights would be blue and light up in the direction of the cobot movement. The participants again identified colour-blindness as a limitation, and that the indicator cannot be seen from all directions in the current design, though this was mentioned as an easy problem to fix.

6.3 Refinement

Following the workshop, the designs were refined and translated into the same design language. The refined designs one through four can be seen in Figure 6.2. Design 1 was kept largely the same as the workshop concept, but it now indicates that an information screen would be available for all the signalling states besides the default. This was because each of those states require human action, the nature of which cannot necessarily be gleaned from the light signal on the cobot. The screen would inform the intervening worker what needs to be done if they are not familiar with the cobot task. The "optional sound" aspect was not kept as how or for which states it should be implemented was not specified by participants. For Design 2, only the communication elements were included: the screen and the verbal speech. It was chosen to illustrate a state proclamation and a request as participants considered these as important aspects to signal. Design 3 was changed for clarity, a specific scenario to illustrate the difference between what participants meant as predictable vs unpredictable behaviour was created. The illustration was inspired by previous work on robot movement legibility (Dragan et al., 2013). Design 4 was only minimally changed, the refined illustration did not explicitly indicate that the green progress bar also signalled that the cobot was active. This was to ensure that it would be understood that the main purpose of the indicator is for progress. The underlying signal that the cobot was active when the progress bar was green was instead explained in the accompanying description. Indeed, accompanying descriptions for each design were prepared, which included the scenario of use and an explanation of the elements of the design. These descriptions, including their Swedish versions, can be found in Appendix 11.3. An additional design from previous research was also prepared to be included in the set of concepts presented in the expert evaluation. It was a previously proposed solution for cobot communication in industry that had shown positive results (Scholz et al., 2025). It was included to serve as a reference point for the evaluation of the workshop-created designs.

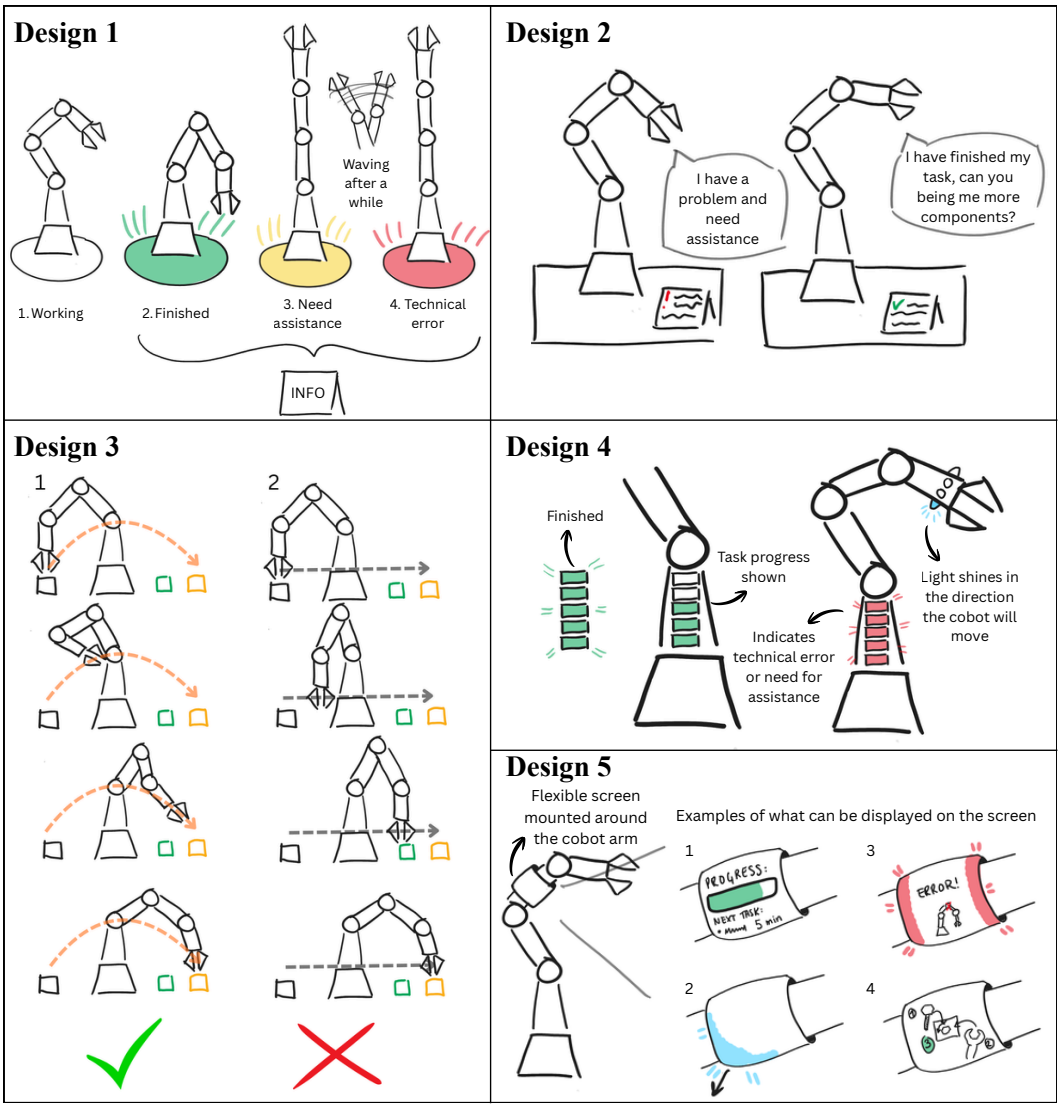


Figure 6.2: Refined designs to be presented to expert evaluators.

7 Expert Evaluation

The second phase evaluated the refined design concepts with relevant stakeholders. The overall goal was to catch design flaws and get context-specific feedback. It thus serves as a basis for further developing the most promising designs or aspects of them.

7.1 Method

The design concepts were presented and feedback was received through an online questionnaire sent out to participating company contacts.

7.1.1 Participants

The questionnaire had eight respondents. The respondents all had experience with industrial settings, either through production management, being production or human factors engineers, industrial technology developers or industrial designers. Demographic information was not collected with the aim to receive more honest answers by protecting the anonymity of the respondents. The primary contacts at participating companies were obtained through university and personal contacts. Snowball sampling was also utilised, as the primary contacts were asked to forward the questionnaire to one or two relevant individuals.

7.1.2 Materials

The questionnaire had three main parts, introduction, design evaluation, and ending. The introduction included brief description of the nature, structure and purpose of the questionnaire. Then short information about industrial cobots was given, including a definition and description, their distinction from other industrial robots, and their features allowing for close human collaboration. Additionally, an image of example cobot applications was shown (Patil et al., 2023). Contact details for the researchers were also provided. Consent information was then provided, and the participants could choose whether to consent and proceed, or not and exit (consent information in Appendix 11.4).

The design evaluation section started with instructions stating that the participants will be presented with the designs and scenario descriptions one at a time, and that for each they

should imagine that design would be implemented in their workplace and answer the questions accordingly. The instructions finished by stating there are no right or wrong answers, and that we are appreciative of any feedback they might have.

After the instructions, the design concepts (Figure 6.2), their descriptions, and accompanying questions were presented in order from Design 1 to Design 5. The accompanying questions for each design were: "Which strengths and weaknesses can you see with this design?", "Does this design communicate information that is relevant for tasks performed in your workplace", and "Is there anything that should be communicated in this scenario, but that cannot be communicated with this design?".

At the end of the questionnaire the participants were thanked for their participation and invited to leave any feedback or thoughts they may have in a free-text answer.

7.1.3 Analysis

A qualitative content analysis was conducted, where participant statements were organised into the categories of: strengths, weaknesses, and suggestions. The frequency of each statement was also counted. (Hsieh & Shannon, 2005). Additionally, the second question (regarding relevance of communicated information) responses were tallied into positive, negative, or missing responses. The statements in the categories, their frequencies, and the second question tally were used to determine which designs to further develop and bring into the validation stage. The weaknesses and suggestions were then used to improve the designs based on the expert feedback. The feedback was summarised and translated from Swedish by the author wherever applicable.

7.2 Results

A summary including the frequency of each statement can be seen in Table 7.1, but the feedback is also described below.

For Design 1, where cobot states were signalled through colours on the baseplate and gestures, positive feedback included that the information was clear and simple, as well as that the design was appropriate for a noisy context. Weaknesses identified were that more detailed information cannot be seen, the waving is not effective and could be misinterpreted as regular movement, workers cannot perceive signals well when far away or out of line of sight of the cobot, and that not all states are distinguishable for colour-blind individuals. Suggestions included that the screen should have information about each scenario (including when cobot is working normally) and could include action based information for the worker, industry standard colours should be used (green for normal operation, blue for missing materials), flashing of the lights can be used for more attention, and alternative ways of escalation include through audio, blinking or elevated lights. Five participants said the information displayed is relevant for their workplace, and the rest did not answer.

Table 7.1: Qualitative content analysis of expert evaluations for each design. Categories include strengths, weaknesses and suggestions, and frequencies for statements within are presented. Frequency counts for answers to the question regarding whether the information communicated is relevant at the participant's workplace is shown at the bottom of each design.

Strengths

DESIGN 1		DESIGN 2		DESIGN 3		DESIGN 4		DESIGN 5	
Frequency		Frequency		Frequency		Frequency		Frequency	
6	Clear or simple information	3	Immediate clear follow up on the screen	2	Worker can plan work tasks due to clear and predictable movement	5	Good, clear or relevant information	5	Can display detailed, clear and relevant information
1	Good for noisy environment	3	Can alert and explain states	1	May lead to less accidents	1	Good for noisy environment	1	Good to have information in same place
		2	Can make for fun and interactive conversation			1	Allows worker to assess when they will be needed again	1	Good to indicate upcoming movement, enhances safety
		2	Good to detect human			1	Recognisable progress indicator		
						1	Lights to warn about upcoming movement is good		

Weaknesses

Frequency		Frequency		Frequency		Frequency		Frequency	
3	More detailed information not displayed	5	A lot of potential idling time, due to not having a way to signal when human is away	4	Adds costs in terms of time, energy, efficiency	2	Bracelet is neither good or relevant. Cannot see when light is shining the other way, and with the cobot moving a lot the light would be hard to follow	4	Difficult to see: need to get close, hard to read if its moving, potential glare
2	Worker cannot gather information out of the view of the cobot particularly since lights are only low down	2	Not good for noisy environment	3	Unnecessary as operators learn cobot behaviour	1	How easily progress can be seen highly dependent on cobot placement	2	Too much or too detailed information could create uncertainty
2	Waving is not good. Can be interpreted as normal movement	1	Screen easily overlooked	1	Weak form of communicating intent	1	Only differentiating the finished and error states through colour		
1	Difficult for colour blind individuals	1	Could negatively impact cognitive load						
		1	Talking may not be appreciated by all workers						

Suggestions

Frequency		Frequency		Frequency		Frequency		Frequency	
3	Screen should have information in all scenarios and indicate what action human should take	2	Have status indication as in Design 1	1	If human needs to place something in a certain location for the cobot to work, this should be indicated clearly	1	More of the robot's state should be communicated, aspects relevant to the task	3	Have a screen a bit away from the cobot or portable
2	Use industry standard colours, green = working, blue = missing material	1	Can have light signals to grab the attention, and then explain and show on the screen when worker acknowledges and is nearby			1	Different indication for system status that does not overlap with progress bar	2	Short information
1	Add flashing for when cobot needs assistance					1	Maybe blinking white instead of blue		
1	As opposed to waving, escalation could be done through e.g. audio, blinking, elevated lights					1	Better with separate lights or screen that can be strategically placed for operator viewing		

Tally for question 2 (information relevance)

Positive (caveat)	Negative	NA	Positive (caveat)	Negative	NA	Positive (caveat)	Negative	NA	Positive (caveat)	Negative	NA	Positive (caveat)	Negative	NA
5	0	3	2 (3)	1	2	2 (1)	1	4	5 (1)	0	2	2 (2)	1	3

Design 2, with verbal communication and supportive screen, had the positive feedback that it might be fun and interactive, that its good to detect when the human is nearby, that there is possibility for immediate follow-up information on the screen, and that it can alert and explain states. Weaknesses identified were that there is a lot of potential idle time, that speech is not appropriate for noisy environments, that the screen may be easily overlooked, that the workers' cognitive load could be negatively impacted, and that not all workers might appreciate being talked to by a cobot. Suggestions included to have status indications like in Design 1, and that attention could be called through light signals if human is not nearby and then the cobot follows up with explanation when the human is nearby and has acknowledged the signal. Two participants said the information is relevant, and three said it could be (if worker was nearby, and with relevant information content). One participant said the design would not work due to noise interference and two participants did not answer.

Design 3, with predictable movement, had the positive feedback that the worker can plan tasks better due to the clear and predictable movement, and that it may improve safety. Weaknesses include added costs (energy, time, efficiency), that the solution is unnecessary because the worker learns cobot movement regardless, and that it is a weak form of communicating intent. Suggestions included that if the human's task includes placing material at a specific location for the cobot, this should be indicated clearly. Two people said the information would be relevant, one said it would probably be relevant, one said no, and four did not answer.

Design 4, with the integrated progress bar and light band, had the positive feedback that the information is good, clear, and relevant, it is suitable for noisy environments, had a recognisable process indicator, it lets the worker assess when they will be needed again, and it warns before movement. Weaknesses identified were that the bracelet is not useful as it would be hard to follow in movement and sometimes not visible as it can be behind the arm, that the progress bar visibility depends on the cobot placement, and that states are not distinguishable for colour-blind individuals. Suggestions were that more of the cobot state should be communicated, that state indication should not overlap with the progress bar, that the light band should be blinking light, and that it would be better to have separately placed lights in a strategic place for operator viewing. Five participants indicated the information was relevant (though one participant did not like specifically the light band), one said probably, and two did not answer.

Design 5, with the flexible screen around the cobot (Scholz et al., 2025), received the positive feedback that it can display detailed, clear and relevant information, that indicating upcoming movement can be safety enhancing, and that information is collected in one place. Weaknesses identified were that too much or too detailed information could create uncertainty, and that it may be difficult to see, mainly when in movement. Suggestions included having the screen separate from the cobot, or for it to be portable, and that the information displayed should be concise. Two participants said the information was relevant, one said probably but with the risk of too much information, one said it would be useful in assembly applications but not in welding or painting, one said it was not relevant, and three did not answer.

7.2.1 Summary

Design 1 and 4 were the overall best performing designs for three main reasons. First, they have the highest relevance scores based on the second question tally. Neither of the designs had any participants claiming it communicated irrelevant information for their workplace. Additionally, Design 1 had five positive relevance responses and Design 4 had five positive, and one positive with a caveat, which outperforms any of the other designs. Second, strengths were frequently identified by the participants such as the information being good and clear. Other designs also had participants identify such strengths, however, those designs had more critical weaknesses. Indeed, the third reason is that Design 1 and 4 have more actionable weaknesses and suggestions for further iteration, as discussed in the next section.

Design 2 does not work well in a noisy environment and would have high idle time, Design 3 adds costs and is seen as unnecessary, and Design 5 has issues with being able to effectively see the information.

7.3 Design Development

Based on the content analysis and that Designs 1 and 4 represent independent and simultaneous scenario designs respectively, they were targeted for further development.

Design 1 was adapted to align with industry standard colours, to show specific supporting information on the screen for all states, and to add flashing to the critical state which attracts more attention and has the benefit of disambiguating the states for colour-blind individuals. One identified issue remains, that the human cannot see the cobot status from far away, especially since the lights are at the base which may not be optimal for human eye-level. To solve this but still stay within the same concept, an additional light band around the cobot's arm was added and the base protrudes more from the table. Thus, when it stands up to attract attention, it is easier to spot from afar. Additionally, the waving was not well received and dropped for the final design, though gestures for being more recognisable (standing up) when an issue is encountered and signalling standby (slumping down) when being finished were retained.

For Design 4, the main weaknesses were solved through adding a flexible screen which was inspired by Design 5. This solves the issue of not being able to differentiate states and allows the information to be seen from more angles. The flexible screen can display short rolling text which also allows for more detailed and task relevant information to be displayed. The main weaknesses from Design 5 regarding the difficulty of reading the screen is also solved as it is now mounted on a stationary part of the cobot in the view of the worker. The bracelet also no longer indicates the direction of movement, but only that movement is upcoming, as a sort of warning, which is also why it is now yellow. One participant suggested that the design should instead have strategically placed lights or screen for operator viewing, however, this design is meant for a simultaneous scenario meaning the worker will not be far from the robot. The final designs with these changes can be seen in the storyboards in the 8.1.3 Stimuli section below.

8 Empirical Assessment

The final phase of the study focused on assessing cognitive ergonomics and UX outcomes of the iteratively developed designs with industrial workers.

8.1 Method

The two chosen designs were illustrated through storyboards and evaluated through questionnaires. One questionnaire assessed the independent scenario design and the other the simultaneous scenario design, each administered to separate participants.

Questionnaires were chosen over interviews to better enable quantitative measures of cognitive ergonomics and UX measures. With that said, if access to workers were not as time-limited (directives from industry contacts were maximum 15 minutes), or more time to complete data collection was possible, it would have been ideal to also have qualitative data from worker interviews.

8.1.1 Participants

An a priori power analysis in G*Power 3.1.9.7 indicated a sample size of 35 to detect a medium effect with 80% power and a significance level of 0.05. The final sample was approximately half of this target sample. The demographic information for each questionnaire can be seen in Table 8.1.

All participants were recruited through convenience sampling based on who was at work, available, and willing, on the day the researcher was visiting. Any location-specific recruitment information is detailed in the 8.1.6 Procedure section.

8.1.2 Experimental Design

The questionnaires utilised a counterbalanced within-subjects design. The independent variable was the way the cobot communicated. The independent variable had two conditions, suggested and baseline. The suggested design condition was where the cobot communicates as per the two chosen designs in the previous phase. The baseline condition was where the cobot only signalled through a traditional screen. The independent variable was operationalised through illustrating the cobot communication in storyboards, which are detailed

Table 8.1: Participant demographics for the independent and simultaneous questionnaires in the empirical assessment phase

		Independent	**Simultaneous**
Sample size		14	13
Work area	Processing	12	12
	Assembly	2	1
Years of experience	More than 10 years	9	7
	6–10 years	1	3
	3–5 years	1	3
	Less than 1 year	1	0
Gender	Male	9 (64%)	10 (77%)
	Female	5 (36%)	3 (23%)
Age	Average age	37.6	41.2
	Age range	38	37
Colour-blind		1	0

in the 8.1.3 Stimuli section below. The dependent variables were the same in both questionnaires and included SA, workload, perceived safety, and UX. These are detailed in the 8.1.4 Measures section below.

8.1.3 Stimuli

Storyboards were chosen to depict the designs as the outcome measures are situated in specific situations, but it was not possible to implement the design concepts on physical prototypes. Additionally, storyboards were chosen over free-standing designs (as in the expert evaluation) as clearly demonstrating the independent and simultaneous nature of the scenarios to the participant became possible.

In the independent questionnaire, the scenario illustrated the CNC machine tending application shown in the workshop phase [1]. Thus, participants in this questionnaire see the same context the design was developed in. Additionally, machine tending is one of the most common industrial cobot applications (Universal Robots, 2022), and CNC machines would have been familiar to participants from two of the industrial locations as they were prevalent there. Two storyboards were created, one for the suggested design, and one for the baseline. The baseline design used a regularly mounted screen, which commonly accompanies models of cobots in use and are common in industry for general machine observation. Each storyboard has four frames (Figure 8.1). The first frame describes the cobot task and the operator task, and illustrates spatial relationship between cobot and human in the scenario. The cobot task is to load and unload the CNC machine, whereas the human's task is to carry out their own assembly task and monitor the cobot. Additionally in the suggested design, the frame text states that green means the cobot is working. Such explanations of colours were added to each frame for the suggested design storyboard because in industrial scenarios operators are trained on the technology they use and would in a real scenario know this information. The second, third and fourth frames depict situations where the cobot is out of material, finished, and encountering an error respectively.

In the simultaneous questionnaire, the scenario illustrated in the storyboards was a scenario where the cobot is riveting a metal sheet, at the same time as the worker is screwing screws into the metal sheet. This scenario, with riveting and screwing, was inspired by a generic example of a simultaneous scenario in Keshvarparast et al., 2024. The scenario shown in the workshop phase was too complicated to understand and illustrate, which is why it was opted to use a simpler but still simultaneous scenario. Again, two storyboards were created, one for the suggested design, and one for the baseline design. Each storyboard has five frames

[1] Productive Robotics. (2023, August 14). *Why swiss productions, etogen precision & true precision chose the OB7 cobot*. https://www.youtube.com/watch?v=HrTFCaQdjqU

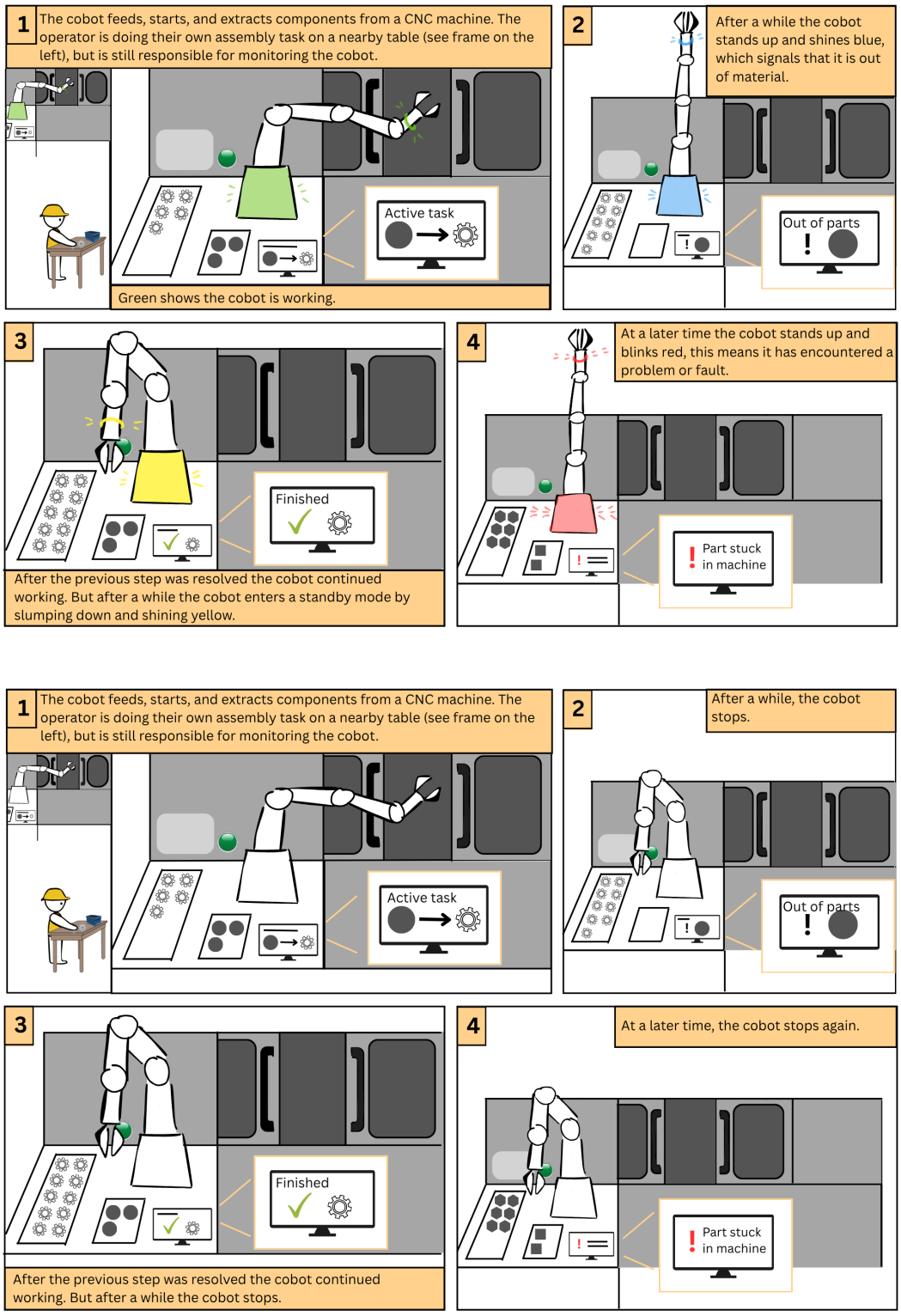


Figure 8.1: Independent stimuli for the suggested (top) and baseline (bottom) designs. The suggested design stimulus is based on the concept generation and expert evaluation phases of the study. These storyboards have been translated into English by the author for the purpose of the thesis. The original storyboards, in Swedish, can be seen in Appendix 11.6.

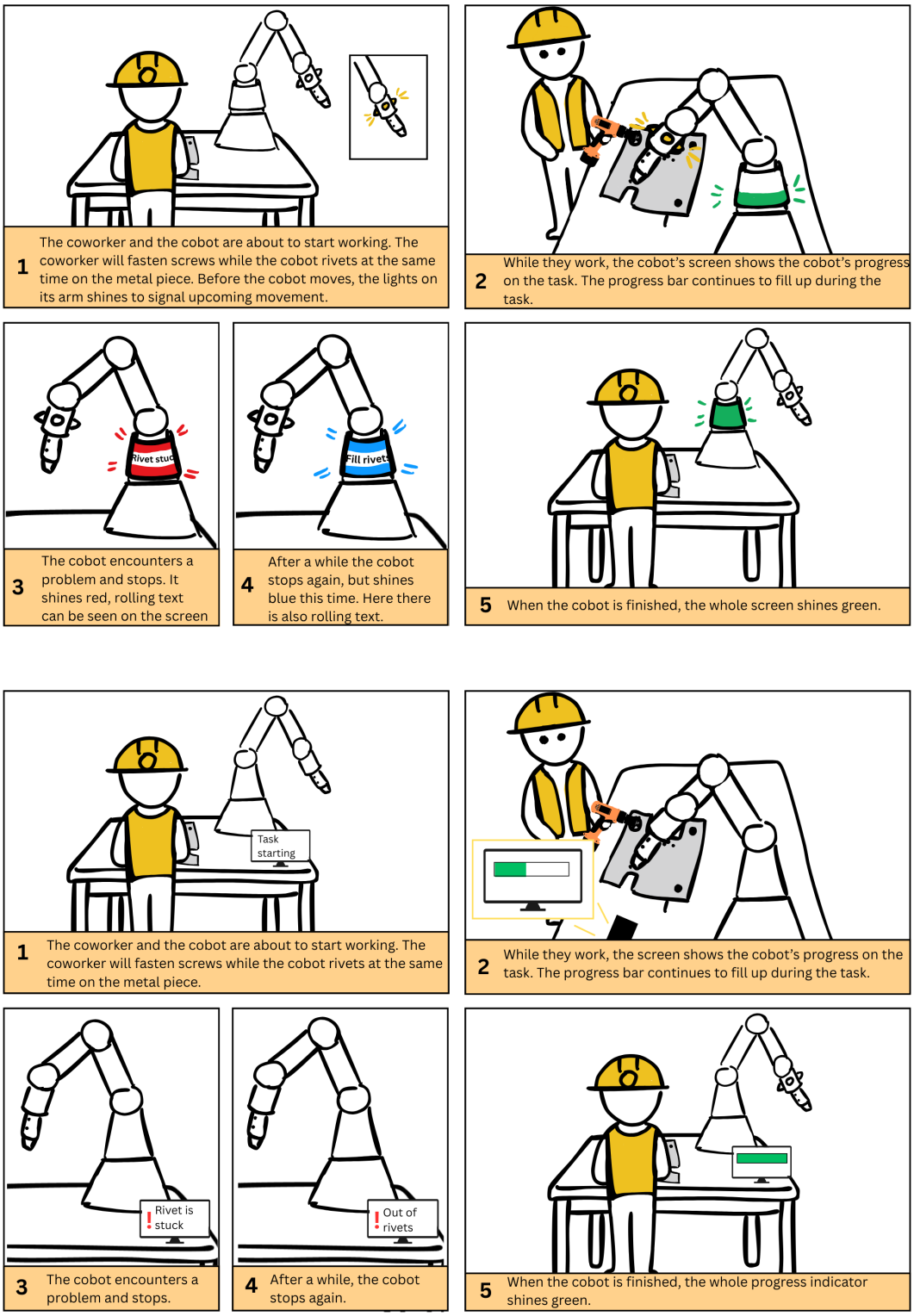


Figure 8.2: Simultaneous stimuli for the suggested (top) and baseline (bottom) designs. The suggested design stimulus is based on the concept generation and expert evaluation phases of the study. These storyboards have been translated into English by the author for the purpose of the thesis. The original storyboards, in Swedish, can be seen in Appendix 11.7.

(Figure 8.2). The first frame again describes the tasks, in this case screwing and riveting. For this storyboard the spatial relationship between the human and the cobot is depicted on several frames. For the suggested design storyboard, the text in the first frame also describes that the lights on the cobot arm light up before movement. The second frame describes that the screen on the cobot shows task progress, and that the bar keeps filling throughout the task. The third and fourth frames depict and describe an error of a rivet being stuck and a missing materials error respectively, both frames also describe that the text on the screen is rolling. The fifth and final frame describes that the whole screen lights up when the task is finished. The baseline storyboard is similar, but without irrelevant text descriptors and with information portrayed on the table mounted screen instead.

8.1.4 Measures

There were four measures in the questionnaires, each is described below.

Situation Awareness

SA was measured through the SA3 (Braarud, 2021), a questionnaire based on the levels of Endsley's (1995) SA theory. The scale has three items, one for each level of observation, comprehension and prediction (also known as projection). Before the items, the instruction to the participant was to answer the questions in relation to the way the cobot signals in the storyboard. Each item was on a 11-point likert scale, where lower scores indicate better perception/comprehension/prediction. The items were translated into Swedish, and rephrased to be about how the participant would observe, comprehend, and predict, since they did not actually experience the scenario.

The SA3 measure was chosen over the SA rating technique (SART) because it relates closely to the theoretical view of SA discussed previously (Endsley, 1995) and is very short, which is crucial due to aforementioned time constraints.

Workload

Workload was measured through the mental effort and psychological stress load items of the Subjective Workload Assessment Technique (SWAT) (Reid & Nygren, 1988). One item was for cognitive workload, and one for psychological stress, each having three levels of increasing workload. The items were translated into Swedish for the questionnaire and rephrased to be in a hypothetical scenario, given participants had not experienced a cobot interaction.

The SWAT was chosen over the NASA-TLX (Hart, 1986) again because of its brevity. Additionally, the NASA-TLX seemed to be more difficult to rephrase into hypotheticals, and several dimensions like temporal demand, effort and physical demand would be very difficult or impossible to assess based on the storyboard format.

Perceived Safety

Perceived safety was assessed through a method used in Akalin et al., 2019, which was cited as a way to measure perceived safety in HRI by Akalin, 2022. The method uses a five-point semantic differential scale with six items, three items are from the perceived safety part of the Godspeed questionnaire (anxious-relaxed, agitated-calm, and quiescent-surprised) (Bartneck et al., 2009) and three developed in Akalin et al., 2019: threatening-safe, uncomfortable-comfortable and predictable-unpredictable. The items were translated into Swedish for the questionnaire.

User Experience

UX was assessed through the short and Swedish version of the user experience questionnaire (UEQ-S) (UEQ Team, 2024). The UEQ-S has eight items on a seven-point semantic differential

scale. The UEQ-S has two subscales, pragmatic and hedonic quality. The pragmatic quality subscale contains the items: obstructive-supportive, complicated-easy, inefficient-efficient, and confusing-clear. The hedonic quality subscale contains the items: boring-exciting, not interesting-interesting, conventional-inventive, and usual-leading edge.

The UEQ-S was chosen over the full UEQ and SUS partly because of its length, and that it would be familiar to the participant due to previous scales using semantic differentials which may accelerate its completion. Additionally, the UEQ-S includes the subscale of hedonic quality, which was of interest to include due to the UX as opposed to usability focus of this thesis.

8.1.5 Questionnaire

The questionnaire was created in LimeSurvey, but was mostly administered by paper (see 8.1.6 Procedure section for details).

The questionnaire started with an introductory section similar to the one in the expert evaluation questionnaire. It contained a brief explanation on cobots and their use in industry, contact details to the researchers, and consent information (consent information in Appendix 11.5).

Background questions and included which industrial area they work in, how long they had worked in industry, their age and gender, and if they are colour-blind.

The next two sections were the storyboards with the suggested and baseline designs and accompanying questions (order of storyboards was counterbalanced). Each section starts with showing the related storyboard and instructs participants to imagine themselves as the operator in the storyboard, and immerse themselves in the scenario. They were then asked to answer the questions that followed.

In each condition there was first a question asking the participant to identify all the ways the cobot is signalling in the storyboard, this was to ensure they had a focus on cobot signalling and paid attention to all relevant aspects. For the independent scenario the options were: cobot base, band around the cobot's arm, screen, and speech. For the simultaneous scenario they were: screen around the cobot, light band around the cobot's arm, screen on the table, and speech.

This was followed by the measures SA, workload, perceived safety, and UX as detailed in the 8.1.4 Measures section above. Between workload and perceived safety there was also a question that asked the participant to identify what their role as operator was in the storyboard. The options were: complete own assembly task, monitor the cobot, and cooperate with the cobot on a mutual task. This question was added to ensure they understood the operator role.

At the end of each condition there was a free-text answer asking if they had any comments about the design of the cobot signals. The question gives examples of what to consider. For the independent scenario it included what their opinion was on the colours, placements of lights and screen, the information communicated, and movements (with redundant aspects removed for the baseline condition question). For the simultaneous scenario it was the screen on the cobot including its placement and information, and the light band around the arm (in the baseline it was instead the placement and information of the screen on the table).

At the end of the questionnaire participants were thanked for their participation and asked their preference between the conditions, an option that they were equal was also provided. The end section also had a question regarding if it was easy to immerse themselves in the storyboard scenario (a short free-text answer). This question was intended to exclude participants who had difficulty imagining the scenario, and one participant was excluded due to finding it particularly difficult. Finally, there was a section where they could leave any other comments they had about the questionnaire (short free-text answer).

8.1.6 Procedure

The procedure for administering the questionnaires differed depending on the possibilities of each participating company location.

At location A, half the questionnaires were administered by the company contact and the researcher walking out into the processing part of the factory and finding clusters of workers willing to participate. The researcher first gave a short introduction to the questionnaire before handing them out. The researcher and company contact then stayed for the duration and answered any questions participants had. The other half of the questionnaires were handed out to workers by the researcher (accompanied by the company contact) and collected two hours later. This was due to other commitments of the contact, and safety regulations preventing the researcher from entering the factory without accompaniment.

At location B, the workers met the researcher in an office, one by one. The participants had been recruited on a previous day by the company contact, the participants were chosen based on who the contact met first and was willing. On the first day of visiting five participants answered on the computer, but this took much longer than the paper versions. On the second day, the other five participants answered on paper. The one-on-one format allowed the researcher to explain the nature of the study, and for the paper versions briefly flip through the upcoming questions (as it was noted several previous participants got stuck on certain questions). Once the questionnaire was finished, the researcher asked if the worker had any comments and took field notes of anything mentioned. Even though some participants did the computer version, all participants saw the paper versions of the scenarios. For the participants answering on the computer, the suggested and baseline design storyboards were labelled and put upside down for the participant to flip when they got to the relevant part. This was so they would not have to scroll up frequently to view the storyboard.

At location C, the workers worked within assembly, meaning they did not have as much possible downtime as the workers from locations A and B. The questionnaires were distributed by two production managers to their employees to fill out when they had time, for example during their break or lunch. This meant the researcher could not explain the questionnaire to participants. 22 questionnaires were distributed, and seven came back, one of which was unfinished, and another where the participant indicated they could not imagine themselves in or understand the scenarios, and two with missing marks in the consent box. As such those were excluded, resulting in the three participants from location C.

8.1.7 Analysis

Given the small sample size, and ordinal data, non-parametric tests were used throughout the analysis. For SA, perceived safety and UX, Wilcoxon signed rank tests were used. For workload, the data was analysed as categorical and a McNemar-Bowker test was used instead.

8.2 Results

This section presents the results of the independent and simultaneous questionnaires. Situation awareness, workload, perceived safety, and user experience outcomes are presented for each.

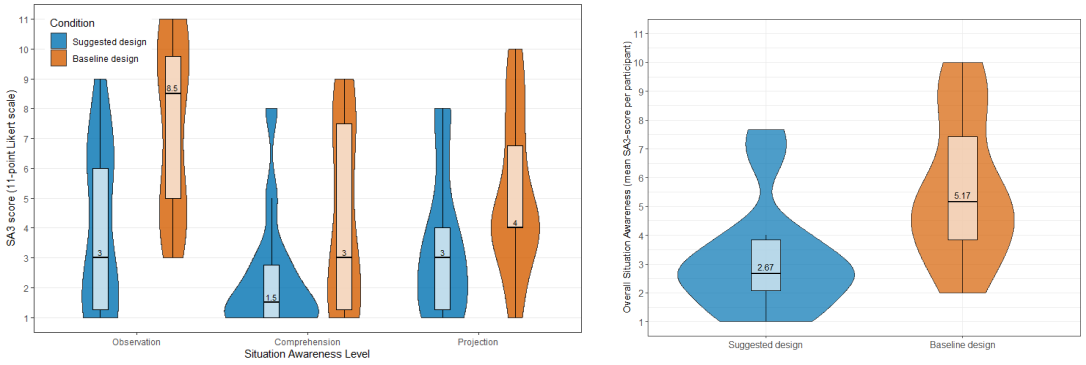
8.2.1 Independent Scenario

Situation Awareness

SA was significantly better for the suggested design compared to the baseline design for each level of SA, see Table 8.2 and Figure 8.3a.

Table 8.2: Wilcoxon signed rank tests and descriptives for comparing suggested and baseline designs for each level of situation awareness (independent scenario). Significant results are in bold, and r is rank-biserial correlation (effect size). (Lower scores indicate better SA)

SA Level	W	p	r	Suggested Median (IQR)	Baseline Median (IQR)
Observation	7.00	**.008**	−0.846	3.00 (4.75)	8.50 (4.75)
Comprehension	2.50	**.020**	−0.889	1.50 (1.75)	3.00 (6.25)
Prediction	6.50	**.020**	−0.803	3.00 (2.75)	4.00 (2.75)



(a) Situation awareness per level and condition.

(b) Overall situation awareness by condition.

Figure 8.3: Box and violin plots of situation awareness for the independent scenario. The left panel shows situation awareness per level (observation, comprehension, and projection) and condition (suggested or baseline design). The right panel shows overall average SA3 score per participant by condition. Lower SA3 scores indicate better situation awareness.

When averaging the subscales, an overall SA index can be obtained (Braarud, 2021). The SA index was better for the suggested design (median = 2.67, IQR = 1.75) compared to the baseline design (median = 5.17, IQR = 3.58) (W = 7.0, p = .005, r = -.867)(Figure 8.3b). A reliability analysis showed that there was an acceptable consistency between the three items both for the suggested (Cronbach's α = .701) and the baseline (Cronbach's α = .821) designs.

Workload

A McNemar-Bowker test showed that there were no significant differences between the suggested and baseline designs for both the mental strain ($\chi^2(3) = 4.8$, p = .200) and the psychological stress ($\chi^2(3) = 3$, p = .400) SWAT subscales. The contingency tables can be seen in Table 8.3.

Table 8.3: SWAT contingency tables for the independent scenario. Values 1-3 indicate levels of mental strain and psychological stress, where lower values represent lower workload. Numbers on the main diagonal represent participants who reported the same rating for both suggested and baseline designs.

Mental Strain

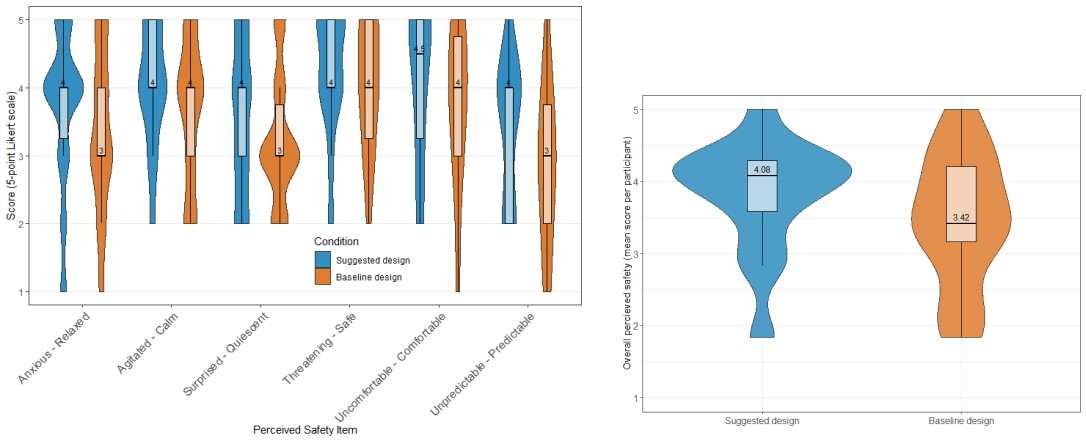
Suggested Design \ Baseline Design	1	2	3
1	3	4	1
2	1	3	2
3	0	0	0

Psychological Stress

Suggested Design \ Baseline Design	1	2	3
1	9	1	1
2	0	2	1
3	0	0	0

Perceived Safety

Akalin et al. (2019) calculated an overall perceived safety score by averaging across the items, and this method was also used here. The overall perceived safety score was significantly better for the suggested design (median = 4.08, IQR = 0.71) compared to the baseline design (median = 3.42, IQR = 1.16) (W = 58, p = .029, r = .758) (Figure 8.4b). A reliability analysis showed that there was an acceptable consistency between all items both for the suggested (Cronbach's α = .866) and the baseline (Cronbach's α = .924) designs.



(a) Perceived safety per item and condition.

(b) Overall perceived safety by condition.

Figure 8.4: Perceived safety results for the independent scenario. The left panel shows perceived safety for each item and condition (suggested and baseline design). The right panel shows the overall perceived safety mean score per participant by condition, with the boxplot indicating the median of those means.

None of the individual items showed a significant difference between the suggested and baseline designs, plots for these items can be seen in Figure 8.4a and their Wilcoxon signed rank tests can be seen in Table 8.4.

Table 8.4: Wilcoxon signed-rank test results for individual perceived safety items (independent scenario)

Item	W Statistic	p
Anxious–Relaxed	13.0	0.168
Agitated–Calm	16.0	0.279
Surprised–Quiescent	20.5	0.298
Threatening–Safe	22.0	0.608
Uncomfortable–Comfortable	29.0	0.124
Unpredictable–Predictable	18.5	0.106

User Experience

The perceived UX was better for the suggested design (median = 5.69, IQR = 1.31) compared to the baseline design (median = 4.13, IQR = 1.97) (W = 77, p = .003, r = .974) (Figure 8.5). A reliability analysis showed that there was an acceptable consistency between the eight items both for the suggested (Cronbach's α = .814) and baseline (Cronbach's α = .920) designs.

The handbook for the UEQ (Schrepp, 2023) suggests to be cautious when interpreting the overall UEQ-S scale since the two subscales could have varying importance for the participants. Thus, each subscale was also analysed. The suggested design had better perceived UX

than the baseline design for both the pragmatic and hedonic quality subscales, see Table 8.5 and Figure 8.5.

Table 8.5: Wilcoxon signed rank tests, descriptives, and Cronbach's α for pragmatic and hedonic quality (UEQ-S) comparing suggested and baseline designs (independent scenario)

				Median (IQR)		α	
Quality	W	p	r	Suggested	Baseline	Suggested	Baseline
Pragmatic	66.00	**.004**	1.000	6.25 (0.88)	5.00 (2.19)	.783	.928
Hedonic	74.00	**.007**	.897	5.50 (2.13)	3.63 (2.25)	.915	.931

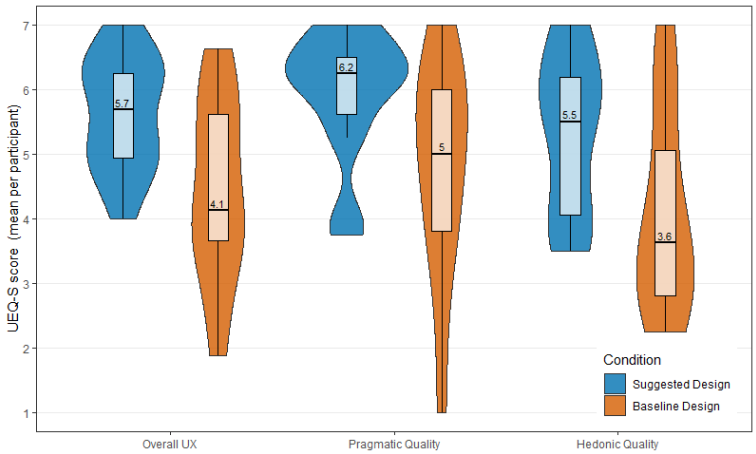


Figure 8.5: UEQ-S scores for the overall scale and the pragmatic and hedonic quality subscales (independent scenario). The y-axis shows mean scores of the items in the scale/subscale per participant. The boxplot shows a median of those means.

8.2.2 Simultaneous

Situation Awareness

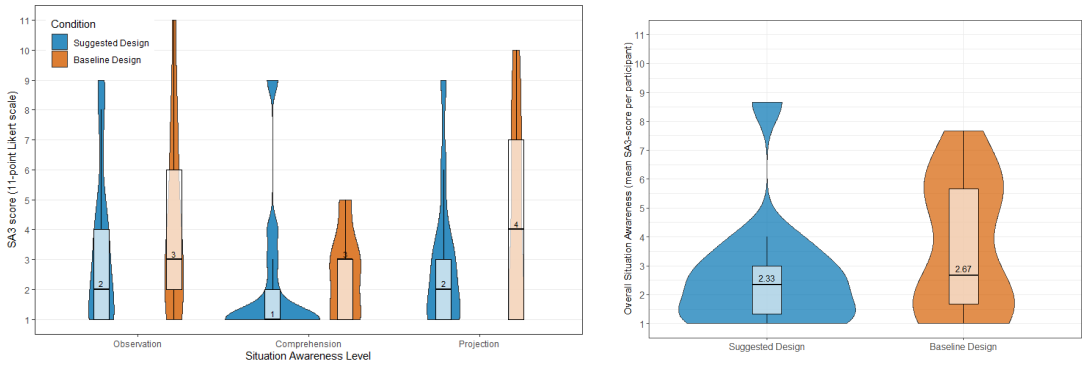
SA did not differ between suggested and baseline designs on any of the SA levels, see Table 8.6 and Figure 8.6a. There was also no difference on the SA Index (suggested median = 2.33, IQR = 1.67, baseline median = 2.67, IQR = 4.00. W = 18.5, p = .213) (Figure 8.6b).

Table 8.6: Wilcoxon signed rank tests and descriptives for comparing suggested and baseline designs for each level of SA (simultaneous scenario). (Lower scores indicate better SA)

SA Level	W	p	Suggested Median (IQR)	Baseline Median (IQR)
Observation	18.0	.357	2.00 (3.00)	3.00 (4.00)
Comprehension	7.0	.268	1.00 (1.00)	3.00 (2.00)
Prediction	7.5	.083	2.00 (2.00)	4.00 (6.00)

Workload

Due to zero counts in some of the paired off-diagonal cells in the contingency table (e.g. no cases in both 3,1 and 1,3 cells), a McNemar-Bowker test could not be successfully performed for either subscale. From a descriptive standpoint, the contingency tables indicate low degree of change and no meaningful difference in SWAT ratings between the designs. The contingency tables can be seen in Table 8.7.



(a) Situation awareness per level and condition.

(b) Overall situation awareness by condition.

Figure 8.6: Box and violin plots of situation awareness for the simultaneous scenario. The left panel shows situation awareness per level (observation, comprehension, and projection) and condition (suggested or baseline design). The right panel shows overall average SA3 score per participant by condition. Lower SA3 scores indicate better situation awareness.

Table 8.7: SWAT contingency tables for the simultaneous scenario. Values 1-3 indicate levels of mental strain and psychological stress, where lower values represent lower workload. Numbers on the main diagonal represent participants who reported the same rating for both suggested and baseline designs.

Mental Strain

Suggested Design \ Baseline Design	1	2	3
1	6	3	0
2	1	1	1
3	0	1	0

Psychological Stress

Suggested Design \ Baseline Design	1	2	3
1	8	2	0
2	1	2	0
3	0	0	0

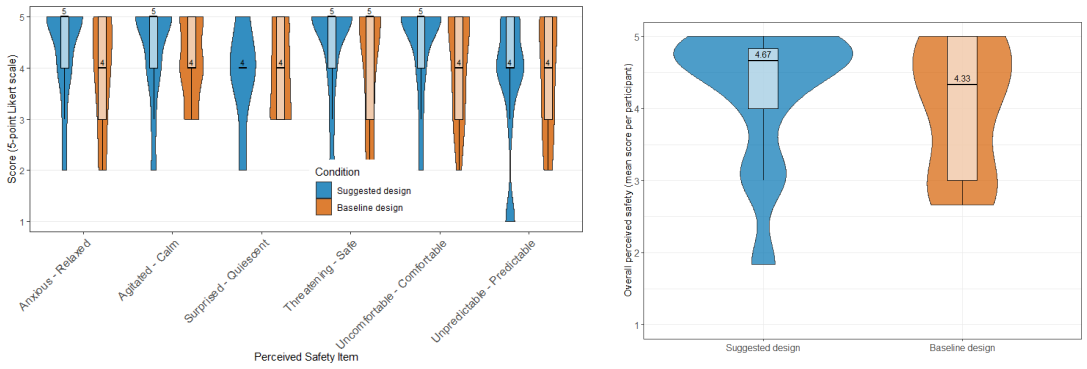
Perceived Safety

The overall perceived safety score did not differ between the suggested design (median = 4.67, IQR = .83) compared to the baseline design (median = 4.33, IQR = 2.00) (W = 35, p = .154) (Figure 8.7b). A reliability analysis showed that there was an acceptable consistency between all items both for the suggested (Cronbach's α = .969) and baseline (Cronbach's α = .946) designs.

The suggested design performed better than the baseline design for the individual items of threatening-safe (suggested median = 5.00 and IQR = 1.00, baseline median = 5.00 and IQR = 2.00, W = 15, p = .037, r = 1.000) and uncomfortable-comfortable (suggested median = 5.00 and IQR = 1.00, baseline median = 4.00 and IQR = 2.00, W = 15, p = .037, r = 1.000) (Figure 8.7a).

User Experience

UX did not differ between the suggested and baseline designs overall or on subscales, though reliability was acceptable, see Table 8.8 and Figure 8.8.



(a) Perceived safety per item and condition.

(b) Overall perceived safety by condition.

Figure 8.7: Perceived safety results for the simultaneous scenario. The left panel shows perceived safety for each item and condition (suggested and baseline design). The right panel shows the overall perceived safety mean score per participant by condition, with the boxplot indicating the median of those means.

Table 8.8: Wilcoxon signed rank tests, descriptives, and Cronbach's α for user experience (UEQ-S) comparing suggested and baseline designs

			Median (IQR)		α	
Quality	W	p	Suggested	Baseline	Suggested	Baseline
Overall UX	56.00	.195	5.63 (0.75)	5.25 (1.87)	.928	.940
Pragmatic	34.00	.192	6.50 (1.00)	5.75 (1.75)	.990	.890
Hedonic	53.00	.286	5.00 (0.75)	4.50 (2.00)	.832	.924

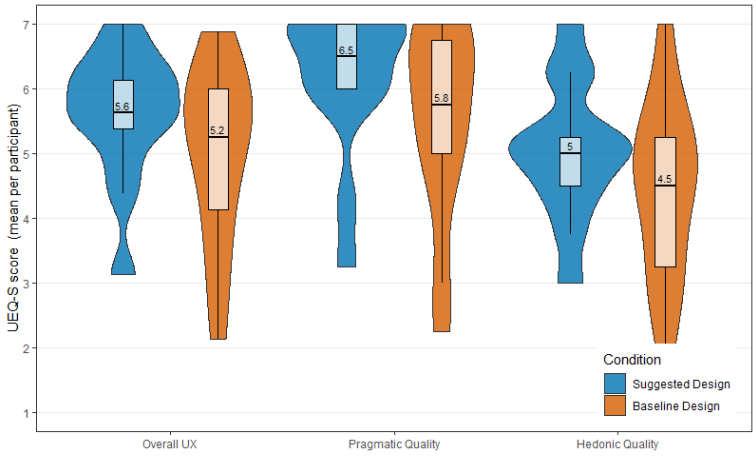


Figure 8.8: UEQ-S scores for the overall scale and the pragmatic and hedonic quality subscales (simultaneous scenario). The y-axis shows mean scores of the items in the scale/subscale per participant. The boxplot shows a median of those means.

9 Discussion

The aim of the thesis was to investigate how cobot-to-worker communication can be designed to support cognitive ergonomics in industrial HRC. This was achieved through developing designs with relevant stakeholders and testing the designs with industrial workers. Two scenario-specific designs were developed, a design signalling cobot state through lights for an independent scenario, and a design signalling cobot state, progress and movement through an integrated screen and lights for a simultaneous scenario. These concepts were presented through storyboards to industrial workers.

For the independent scenario, results showed that industrial workers rated their SA, perceived safety, and UX higher, while cognitive workload remained stable, for the developed design compared to a baseline. This suggests that the design concept has the potential to improve cognitive ergonomics and UX for the workers. Given the increase in SA, it is also possible the suggested design was more intuitively transparent to the worker.

Interestingly, these cognitive ergonomic and UX improvements were not found in the simultaneous scenario. It is therefore possible that more aspects of transparency information are needed to better support the worker in more interactive scenarios.

9.1 Designing for Cognitive Ergonomics

Several aspects of the developed independent scenario design may have contributed to cognitive ergonomic support, including attention, clarity and modality.

First, the design supported attention attraction since it provides salient and valuable information while the effort to seek the information remains low (Wickens & Carswell, 2021). Salience in the design was mainly achieved through the large, coloured light signals and movement, which also contributes to low effort signal detection as it improves the ability to perceive signals at a glance. Cobot state is valuable information for the worker, which is both confirmed by the expert evaluations and previous research (e.g. Arntz et al., 2020; Tang et al., 2019).

Secondly, an aspect of attention is divided attention, and making the signals in the design clear and easy to understand can contribute to more successful congruent processing of multiple tasks, such as progressing one's own task while monitoring the cobot as in the independent scenario (Wickens & Carswell, 2021). Creating clear signals was specifically considered by the participants of the second workshop to support the need for divided atten-

tion. Indeed, both expert evaluators and several participants in the empirical questionnaire indicated that the signals in the developed designs were clear.

Finally, the design relies on the visual modality which can be beneficial in the industrial environment as sound signals get more perceptual and cognitive interference (Baddeley, 2000; Baddeley & Hitch, 1974; Bejerano et al., 2018; Gross & Krenn, 2024; Nielsen et al., 2026). Indeed, in the expert evaluation this was a concern for the verbal speech-based design. However, generally combining modalities can be beneficial (Burke et al., 2006; Cao et al., 2023; Villani et al., 2018), and a participant in the empirical assessment specifically mentioned in a free-text answer that an additional auditory signal would be necessary for them for full monitoring. Also, during the second workshop sound signals were briefly considered as an optional feature, but since this was not further developed it was not brought into the refined designs. The lack of multimodal signals is a potential limitation of the design concepts in this thesis, and something that future research could investigate.

Interestingly, the design concept also follows some of the cobot design guidelines validated by Gualtieri et al., 2022. Followed guidelines include to inform worker about cobot state, provide intuitive communication, avoid human-like features of the cobot, only provide information when relevant and necessary, and avoid overloading worker with feedback. The guideline to avoid human-like features was not actively considered or investigated in this study. Previous research is divided with some arguing for more social cue integration (Cao et al., 2023; Crivelli & Balconi, 2025) and some against it (Bergman et al., 2019; Nielsen et al., 2026). The current study does not aim to take a stance but does indicate that cognitive ergonomic and UX benefits can be achieved without relying on anthropomorphic features or social cues.

Gualtieri et al., 2022 found that workstation designs that implemented more of their developed guidelines improved, for example, usability, perceived enjoyment, and cognitive workload. In the present study, the independent baseline condition followed less of these guidelines by having less intuitive communication and not having the option to seek out more information only when relevant (lights in the suggested design mean workers seek more information when needed as opposed to intermittently). Thus, adherence to these guidelines in the suggested design may have played a role in the positive results seen in the independent scenario. As a note, in the independent scenario, the cognitive workload did not decrease, but it also did not increase with the suggested designs. This is positive in the sense that adding more signals (lights in addition to the screen) did not lead workers to perceive higher cognitive workload. However, workload may have been generally low in this type of scenario, and perceived workload may have been low for participants as a majority had more than ten years of industrial experience. This is relevant as expertise can help mitigate the cognitive workload required to acquire SA (Endsley, 2021). Future investigations in more high workload scenarios, and with participants with more varied experience may provide better insights into true effects on workload.

Still, the aforementioned guidelines need to be further developed. They are general in nature, and do not provide specific guidelines for interface and communication design. For instance, they suggest providing intuitive communication, and show one way of doing that in their selected industrial scenario (Gualtieri et al., 2022). However, they do not provide more specific information on what it means to have intuitive communication or how it is best implemented. Thus, the guidelines can be supplemented with interface specific guidelines, perhaps divided by industrial scenario.

As a general note, the developed design concepts have similarities to previous research, such as utilising LED lights around the cobot (Tang et al., 2019), similar gestures (slumping down) (Arntz et al., 2020; Takayama et al., 2011), communicating cobot states through coloured lights (Cao et al., 2023, 2024; Hoecherl et al., 2018), having supplementary screens (Aubert et al., 2018; Hoecherl et al., 2018), and warning for movement (Arntz et al., 2020; Scholz et al., 2025). This suggests that successful solutions can sometimes carry over across scenarios. Additionally, given how positively the expert evaluators viewed these features, it

suggests alignment between real-world desires for the industrial context and academically investigated solutions. However, an interesting discrepancy between the expert evaluators and previous research is in terms of legible or predictable motion. While predictable motion has been shown in previous research to be beneficial (Aubert et al., 2018; Dragan & Srinivasa, 2014), the expert evaluators instead suggested that it is unnecessary as workers will learn cobot movement regardless. It is possible this view comes from a lack of focus on more dynamic situations (i.e. experts considering tasks with specific and repetitive component picking-orders, which is indicated by their responses). Though, these experts are not themselves workers, and workers may have different views. Still, the overall sentiment contrasts with previous research that emphasises adapting cobot behaviour to the worker rather than expecting the worker to adapt to the cobot.

In summary, the design of the RtH communication in the independent scenario indicates positive effects on perceived cognitive ergonomics of the worker, but the suggested design still has limitations, as will be further showed in the next section. Additionally, limitations of previous design guidelines also exist and should be further developed and specified.

9.2 Cobot Transparency in Different Scenarios

Several of the outcomes of this study can be understood in terms of the SAT model (Chen et al., 2014). Based on the design concepts that were generated, and the comments from the expert evaluators, it seems the information signalled and desired is mainly on the first level of the model. Level 1 suggests the operator should be provided with information about state, goals, intentions and proposed actions (Chen et al., 2014). States being the most prominent communicated aspect of both developed designs, but intentions in terms of movement was also included. In the independent scenario, expert evaluators wished to have the current cobot task also displayed, which it was in the final design (an aspect that is also noted as important information in other previous works (Cha et al., 2018; El Zaatari et al., 2019). This was a way of displaying the goal, as it shows which parts are being processed into which. The third level, about projections of future states, can be interpreted as slightly represented in the simultaneous scenario through the progress bar and upcoming movement signal. The progress bar can help project the future state of the system through indicating time left until completed task. Though it should be noted that while level 3 of SAT is agent's projection of future states, the examples provided by Chen et al. (2014, p. 3) include predicted consequences or probability of success, which is not represented in either design. Still, depending on the progress bar indication, workers can understand if the competed task state is near or far from the current situation. While a relatively weak form of projection, the light band also warns the worker about an upcoming state, which in this case is movement. The lack of representation of level 2 and 3 information can be seen as a limitation of both designs. While not each level is strictly necessary for achieving transparency, the authors suggest that incorporating each level provides good transparency (Chen et al., 2014). Thus, the design concepts could be further improved through adding information suggested on the second and third level.

Interestingly, since level 3 is represented in the simultaneous scenario, but not the independent, there may be an inclination towards stating the simultaneous scenario is more transparent. Despite this (and having similar design intents to the independent scenario in the previous section), the simultaneous scenario showed no significant differences on cognitive ergonomics outcomes. The lack of cognitive ergonomic improvement suggests more (or different) transparency information may be needed in this more interactive cobot scenario (though this result could be due to methodological limitations as discussed in the 9.3 Methodological limitations section below). A general implication for the frameworks use in cobot applications is that the transparency requirement may be different for different industrial scenarios. For instance, level 2 of the SAT includes both showing the agents reasoning for

their proposed actions, and constraints they considers in planning them (Chen et al., 2014). The need for this information differs between industrial scenarios. In an independent scenario, the cobot would not need to present proposed actions and the reasoning behind them, this would require too much attention from the worker and be inefficient. However, in a sequential or supportive scenario, this information may be more relevant, for instance the cobot suggesting it should put the workpiece down to refill an empty component container. The SAT model therefore may serve as a framework for determining what levels of SAT and which content should be prioritised in different cobot scenarios.

Overall, the utility of SAT in cobot application is twofold, serving both as a tool to analyse and consider the information included in communication interfaces during the design process, and as a potential framework to determine level and content of transparency information in cobot scenarios.

9.3 Methodological Limitations

The study has some methodological shortcomings including low power, suboptimal cognitive workload measure, unclear simultaneous storyboard, and less than ideal stakeholder involvement. First, the a priori power analysis suggested 35 participants, and both questionnaires only reached less than half. Thus, both the large effect sizes for the independent scenario, and insignificant results of the simultaneous scenario could be due to small sample size. Second, the SWAT measure used for workload is less sensitive at low levels of workload (Hancock et al., 2021) (levels may have been low due to expertise and in the scenario as previously mentioned). This could affect both the independent and simultaneous results, but especially the simultaneous given that, across both subscales, the highest level of workload was only indicated twice and the lowest level was indicated the majority of the time. Another issue particularly relevant in the simultaneous scenario is that only four participants indicated ease of imagining the scenario, and nine did not answer the question (whereas in the independent scenario nine had acceptable immersion and five did not answer). This makes it difficult to assess whether the storyboard facilitated immersion in the scenario. The simultaneous storyboard may also have been generally more unclear due to the scenario being unfamiliar to the workers, leaving more aspects up to interpretation. This is plausible as one participant commented that the baseline screen was in a better location than the cobot-mounted screen, but another participant indicated the baseline screen was at the wrong angle to see it. Additionally, the understanding operator task and signalling channels questions varied considerably across participants for both independent and simultaneous scenarios. This would indicate ambiguity in the storyboards, and it is possible the understanding of the simultaneous scenario was particularly affected due to its novelty. The novelty, low workload, potentially low immersion, and unclarity are all shortcomings that disproportionally affect the simultaneous scenario and could have contributed to insignificant results. A future improvement could be to adapt a current task that workers are familiar with to a simultaneous scenario with a cobot to anchor it better for participants.

Another limitation is that stakeholders and workers could not be involved in the workshops. For future workshops and participatory design it is ideal to have actual users and stakeholders involved in coming up with design concepts and ideas. While there was qualitative input especially from experts through an online questionnaire, it may have been more fruitful to gather experts in groups and conduct full co-design workshops as in previous research (Cao et al., 2024; Scholz et al., 2025). A related limitation is that free-text answers from the workers in the final questionnaire were not analysed due to time constraints and scope of the thesis.

Further, while the scenarios utilised are realistic in terms of possible industrial cobot applications, they were still presented through storyboards. This means the ecological validity is lacking since the participants did not experience a real enactment or physical prototype

of the designs before answering the questions. This is acceptable given the study concerns the early design process, and is a pre-study as opposed to a final validation. Still, it means cognitive outcome results should not be overstated, they are perceptions and preliminary.

In summary, the results of this study should be interpreted with caution. Effect sizes may be overestimated in the independent scenario, and the insignificant results in the simultaneous scenario may not represent a genuine null result but rather be a consequence of methodological limitations.

9.4 Implications

This study contributes to the research on the design for cobot communication in industry. It is the only study, to the authors' knowledge, that had a participatory design approach to RtH communication in an independent and simultaneous scenario. Further, the current study is one of few that evaluated cognitive outcomes with industrial workers early in the design process. The study preliminarily showed that having large, clear, colour-based robot state signals on the cobot is better for cognitive ergonomics and UX of workers in independent scenarios than traditional interface screens. The study also indicates that considering cognitive factors in the design process, especially related to limitations of attention and information processing, can support generation of human-centric design concepts.

In terms of theoretical contributions, the study gives slight indication that transparency information needed for worker SA can vary across industrial scenarios. It highlights both the usefulness of the SAT model, but also the need to tailor RtH communication solutions towards specific cobot applications as opposed to having a universal solution as is currently the standard in cobot manufacturing.

The human-centric focus of this thesis also has broader societal implications for the workers' well-being and the goals of Industry 5.0. The study included outcome measures like UX and perceived safety that are crucial for the well-being of industrial workers. These aspects are a previously understudied aspect of cobot communication and collaboration more generally, with a larger focus having been on improving cobot technology of human-to-robot communication or measuring performance outcomes (El Zaatari et al., 2019; Gross & Krenn, 2024; Hentout et al., 2019). Thus, this study also wishes to advocate for the inclusion of human and cognitive centric considerations and measures when designing and evaluating technological solutions for industrial HRC.

9.5 Future Work

The promising result of the independent design promotes further investigation into the design solution, with more high-fidelity prototyping and testing with users. This could include investigating multimodal signals, in both the independent and simultaneous scenario. It would also generally be of interest to revisit the simultaneous scenario by adapting it into a more familiar work task, and adding more aspects of transparency information as per SAT model recommendations. In fact, to further investigate the SAT application in both scenarios would be interesting to determine if, based on the SAT, the level of transparency needed in cobot communication could be determined for specific tasks. This could lead to a framework of specific and crucial information to be communicated in the name of transparency for the different industrial scenarios of independent, simultaneous, sequential and supportive. Such research could also add more specific and comprehensive guidelines for cobot communication than found in previous research (Gross & Krenn, 2024; Gualtieri et al., 2022). In general, future research on RtH communication in industry should prioritise involving stakeholders in the early design process, given the useful expert contributions in this study. It should also actively measure and consider cognitive ergonomic factors.

10 Conclusion

This thesis aimed to develop and validate design concepts for cobot-to-human signalling in terms of cognitive ergonomic and user experience outcomes. The study investigated two common industrial human-robot collaboration scenarios, independent and simultaneous, but that have been less represented in the literature. Further, the methodological approach included involving stakeholders in the design and validation process to ensure those related to cobot implementation and usage in real industrial settings are not overlooked. In addition to participatory approaches, considering and measuring cognitive ergonomic and user experience factors has ties to Industry 5.0 and aligns with its human-centric perspectives of worker well-being and empowerment.

The results showed that the suggested design for the independent scenario outperformed the baseline on measures of situation awareness, perceived safety and user experience, while not increasing cognitive workload. This suggests the design concept has the potential to support the cognitive ergonomics and overall experience of the human in the interaction. For the design suggested for a simultaneous scenario, there were no differences on any of the outcome measures in comparison to the baseline. While this could be due to methodological limitations, it is also possible the interface is not clear or intuitive enough or does not provide a sufficient level of transparency information to support the cognition and experience of workers. As such, transparency information required may differ from one scenario to another, which highlights the need for application-specific cobot-to-human communication solutions.

Overall, the thesis contributes preliminary insights into how cobot-to-human communication can be designed to support cognitive ergonomics in industrial HRC. It is also hoped the thesis contributes to the human-centric focus of Industry 5.0 by encouraging careful consideration of the workers who will or do operate alongside innovative technologies.

Bibliography

Akalin, N. (2022). *Perceived safety in social human-robot interaction* [Doctoral thesis]. Örebro University. http://urn.kb.se/resolve?urn=urn:nbn:se:hj:diva-66670

Akalin, N., Kristoffersson, A., & Loutfi, A. (2019). Evaluating the sense of safety and security in human - robot interaction with older people. In *Social robots : Technological, societal and ethical aspects of human-robot interaction* (pp. 237–264). School of Innovation, Design; Engineering, Mälardalen University, Västerås, Sweden. https://doi.org/10.1007/978-3-030-17107-0_12

Alves, J., Lima, T. M., & Gaspar, P. D. (2023). Is industry 5.0 a human-centred approach? a systematic review. *Processes*, *11*(1), 193. https://doi.org/10.3390/pr11010193

Anderson, T., & Shattuck, J. (2012). Design-based research: A decade of progress in education research? *Educational Researcher*, *41*(1), 16–25. https://doi.org/10.3102/0013189X11428813

Arai, T., Kato, R., & Fujita, M. (2010). Assessment of operator stress induced by robot collaboration in assembly. *CIRP Annals*, *59*(1), 5–8. https://doi.org/10.1016/j.cirp.2010.03.043

Arntz, A., Eimler, S. C., & Hoppe, H. U. (2020). Augmenting the human-robot communication channel in shared task environments. In A. Nolte, C. Alvarez, R. Hishiyama, I.-A. Chounta, M. J. Rodríguez-Triana, & T. Inoue (Eds.), *Collaboration technologies and social computing* (pp. 20–34). Springer International Publishing. https://doi.org/10.1007/978-3-030-58157-2_2

Arvola, M. (2020). *Interaktionsdesign och UX - om att skapa en god användarupplevelse* (2nd ed.). Studentlitteratur. https://www.studentlitteratur.se/kurslitteratur/teknik/webbutveckling-och-anvandbarhet/interaktionsdesign-och-ux/

Atkinson, R., & Shiffrin, R. (1968). Human memory: A proposed system and its control processes. In K. W. Spence & J. T. Spence (Eds.), *Psychology of learning and motivation* (pp. 89–195, Vol. 2). Academic Press. https://doi.org/10.1016/S0079-7421(08)60422-3

Attahir, M., Chutima, P., & Miralles, C. (2025). Cognitive ergonomics assessment and categorization in human-robot collaborative workplaces: A systematic literature review. *Journal of Industrial and Production Engineering*, *43*(4), 1–40. https://doi.org/10.1080/21681015.2025.2594008

Aubert, M. C., Bader, H., & Hauser, K. (2018). Designing multimodal intent communication strategies for conflict avoidance in industrial human-robot teams [ISSN: 1944-9437].

2018 27th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN), 1018–1025. https://doi.org/10.1109/ROMAN.2018.8525557

Baddeley, A. D. (2000). The episodic buffer: A new component of working memory? *Trends in Cognitive Sciences*, *4*(11), 417–423. https://doi.org/10.1016/S1364-6613(00)01538-2

Baddeley, A. D., & Hitch, G. (1974). Working memory. In G. Bower (Ed.), *Psychology of learning and motivation* (pp. 47–89, Vol. 8). Academic Press. https://doi.org/10.1016/S0079-7421(08)60452-1

Bartneck, C., Kulić, D., Croft, E., & Zoghbi, S. (2009). Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of robots. *International Journal of Social Robotics*, *1*, 71–81. https://doi.org/10.1007/s12369-008-0001-3

Bassi, G., Orso, V., Salcuni, S., & Gamberini, L. (2025). Understanding workers' well-being and cognitive load in human-cobot collaboration: Systematic review. *Journal of Medical Internet Research*, *27*, e75658. https://doi.org/10.2196/75658

Bejerano, G., LeMasurier, G., & Yanco, H. A. (2018). Methods for providing indications of robot intent in collaborative human-robot tasks. *Companion of the 2018 ACM/IEEE International Conference on Human-Robot Interaction*, 65–66. https://doi.org/10.1145/3173386.3177069

Bergman, M., de Joode, E., de Geus, M., & Sturm, J. (2019). Human-cobot teams: Exploring design principles and behaviour models to facilitate the understanding of non-verbal communication from cobots. *Proceedings of the 3rd International Conference on Computer-Human Interaction Research and Applications*, 191–198. https://doi.org/10.5220/0008363201910198

Berti, N., Guidolin, M., Tonello, S., Reggiani, M., & Battini, D. (2025). Cognitive ergonomic challenges in human-cobot assembly: Exploratory laboratory test findings. *IFAC-PapersOnLine*, *59*(10), 2616–2621. https://doi.org/10.1016/j.ifacol.2025.09.440

Braarud, P. Ø. (2021). Investigating the validity of subjective workload rating (NASA TLX) and subjective situation awareness rating (SART) for cognitively complex human–machine work. *International Journal of Industrial Ergonomics*, *86*, 103233. https://doi.org/10.1016/j.ergon.2021.103233

Burke, J. L., Prewett, M. S., Gray, A. A., Yang, L., Stilson, F. R. B., Coovert, M. D., Elliot, L. R., & Redden, E. (2006). Comparing the effects of visual-auditory and visual-tactile feedback on user performance: A meta-analysis. *Proceedings of the 8th international conference on Multimodal interfaces*, 108–117. https://doi.org/10.1145/1180995.1181017

Cai, Y., & Patrick Rau, P.-L. (2026). Neuro-cognitive benefits of high agent transparency on operators' cognitive and affective emergent states in human-agent teaming. *Applied Ergonomics*, *134*, 104738. https://doi.org/10.1016/j.apergo.2026.104738

Cao, H.-L., Elprama, S. A., Scholz, C., Siahaya, P., El Makrini, I., Jacobs, A., Ajoudani, A., & Vanderborght, B. (2024). Designing interaction interface for supportive human-robot collaboration: A co-creation study involving factory employees. *Computers & Industrial Engineering*, *192*, 110208. https://doi.org/10.1016/j.cie.2024.110208

Cao, H.-L., Scholz, C., De Winter, J., Makrini, I. E., & Vanderborght, B. (2023). Investigating the role of multi-modal social cues in human-robot collaboration in industrial settings. *International Journal of Social Robotics*, *15*(7), 1169–1179. https://doi.org/10.1007/s12369-023-01018-9

Cardoso, A., Colim, A., Bicho, E., Braga, A. C., Menozzi, M., & Arezes, P. (2021). Ergonomics and human factors as a requirement to implement safer collaborative robotic workstations: A literature review. *Safety*, *7*(4), 71. https://doi.org/10.3390/safety7040071

Carissoli, C., Negri, L., Bassi, M., Storm, F. A., & Delle Fave, A. (2024). Mental workload and human-robot interaction in collaborative tasks: A scoping review. *International Journal of Human–Computer Interaction*, *40*(20), 6458–6477. https://doi.org/10.1080/10447318.2023.2254639

Cha, E., Dragan, A. D., & Srinivasa, S. S. (2015). Perceived robot capability. *2015 24th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*, 541–548. https://doi.org/10.1109/ROMAN.2015.7333656

Cha, E., Kim, Y., Fong, T., & Matarić, M. J. (2018). A survey of nonverbal signaling methods for non-humanoid robots. *Foundations and Trends in Robotics*, *6*(4), 211–323. https://doi.org/10.1561/2300000057

Chen, J., & Barnes, M. (2021). Human-robot interaction. In *Handbook of human factors and ergonomics* (pp. 1121–1142). John Wiley & Sons, Inc. https://doi.org/10.1002/9781119636113.ch44

Chen, J., Lakhmani, S. G., Stowers, K., Selkowitz, A. R., Wright, J. L., & Barnes, M. (2018). Situation awareness-based agent transparency and human-autonomy teaming effectiveness. *Theoretical Issues in Ergonomics Science*, *19*(3), 259–282. https://doi.org/10.1080/1463922X.2017.1315750

Chen, J., Procci, K., Boyce, M., Wright, J. L., Garcia, A., & Barnes, M. (2014, April 1). *Situation awareness-based agent transparency* (ARL-TR-6905). Army Research Laboratory. Aberdeen Proving Ground, MD. https://apps.dtic.mil/sti/html/trecms/AD1143367/

Chiappe, D. L., Strybel, T. Z., & Vu, K.-P. L. (2012). Mechanisms for the acquisition of situation awareness in situated agents. *Theoretical Issues in Ergonomics Science*, *13*(6), 625–647. https://doi.org/10.1080/1463922X.2011.611267

Clark, A. (2014). *Mindware: An introduction to the philosophy of cognitive science* (2nd ed.). Oxford University Press.

Clark, A., & Chalmers, D. (1998). The extended mind. *Analysis*, *58*(1), 7–19. https://www.jstor.org/stable/3328150

Concetta Manuela, L. F., Antonio, G., Giada, L. S., & Rosa, M. (2025). Toward acceptance of human-robot collaboration in industrial settings: A bibliometric and systematic literature review. *The International Journal of Advanced Manufacturing Technology*, *139*(5), 2139–2160. https://doi.org/10.1007/s00170-025-16039-z

Crivelli, D., & Balconi, M. (2025). Industrial collaborative robotics, humanoid cobots, and social embodiment: A critical look on human–robot interaction. *International Journal of Humanoid Robotics*, *22*(1), 2430001. https://doi.org/10.1142/S0219843624300010

Davies, R. (2015). *Industry 4.0: Digitalisation for productivity and growth* (PE 568.337). European Parliamentary Research Service (EPRS).

De Pace, F., Manuri, F., Sanna, A., & Zappia, D. (2018). An augmented interface to display industrial robot faults. In L. T. De Paolis & P. Bourdot (Eds.), *Augmented reality, virtual reality, and computer graphics* (pp. 403–421). Springer International Publishing. https://doi.org/10.1007/978-3-319-95282-6_30

Dei, C., Meregalli Falerni, M., Cilsal, T., Redaelli, D. F., Lavit Nicora, M., Chiappini, M., Storm, F. A., & Malosio, M. (2025). Design and testing of (a)MICO: A multimodal feedback system to facilitate the interaction between cobot and human operator. *Journal on Multimodal User Interfaces*, *19*(1), 21–36. https://doi.org/10.1007/s12193-024-00444-x

Dragan, A., Lee, K. C., & Srinivasa, S. S. (2013). Legibility and predictability of robot motion [ISSN: 2167-2148]. *2013 8th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 301–308. https://doi.org/10.1109/HRI.2013.6483603

Dragan, A., & Srinivasa, S. (2014). Integrating human observer inferences into robot motion planning. *Autonomous Robots*, *37*(4), 351–368. https://doi.org/10.1007/s10514-014-9408-x

El Zaatari, S., Marei, M., Li, W., & Usman, Z. (2019). Cobot programming for collaborative industrial tasks: An overview. *Robotics and Autonomous Systems*, *116*, 162–180. https://doi.org/10.1016/j.robot.2019.03.003

Endsley, M. R. (1995). Toward a theory of situation awareness in dynamic systems. *Human Factors*, *37*, 85–104.

Endsley, M. R. (2015). Situation awareness misconceptions and misunderstandings. *Journal of Cognitive Engineering and Decision Making*, *9*(1), 4–32. https://doi.org/10.1177/1555343415572631

Endsley, M. R. (2021). Situation awareness. In *Handbook of human factors and ergonomics* (pp. 434–455). John Wiley & Sons, Inc. https://doi.org/10.1002/9781119636113.ch17

European Commission: Directorate-General for Research and Innovation. (2020). *Industry 5.0 – human-centric, sustainable and resilient*. Publications Office. https://doi.org/10.2777/073781

European Commission: Directorate-General for Research and Innovation. (2024). *ERA industrial technologies roadmap on human-centric research and innovation for the manufacturing sector*. Publications Office of the European Union. https://doi.org/doi/10.2777/0266

FANUC America Corporation. (2020, February 7). *Cobots empower employees and assembly line manufacturing at flex-n-gate automotive*. https://www.youtube.com/watch?v=uqdrlXuIEFM

Foglia, L., & Wilson, R. A. (2013). Embodied cognition. *WIREs Cognitive Science*, *4*(3), 319–325. https://doi.org/10.1002/wcs.1226

Fournier, É., Kilgus, D., Landry, A., Hmedan, B., Pellier, D., Fiorino, H., & Jeoffrion, C. (2022). The impacts of human-cobot collaboration on perceived cognitive load and usability during an industrial task: An exploratory experiment. *IISE Transactions on Occupational Ergonomics and Human Factors*, *10*(2), 83–90. https://doi.org/10.1080/24725838.2022.2072021

Giallanza, A., La Scalia, G., Micale, R., & La Fata, C. M. (2024). Occupational health and safety issues in human-robot collaboration: State of the art and open challenges. *Safety Science*, *169*, 106313. https://doi.org/10.1016/j.ssci.2023.106313

Gross, S., & Krenn, B. (2024). A communicative perspective on human–robot collaboration in industry: Mapping communicative modes on collaborative scenarios. *International Journal of Social Robotics*, *16*(6), 1315–1332. https://doi.org/10.1007/s12369-023-00991-5

Gruszka, A., & Nęcka, E. (2017). Limitations of working memory capacity: The cognitive and social consequences. *European Management Journal*, *35*(6), 776–784. https://doi.org/10.1016/j.emj.2017.07.001

Gualtieri, L., Fraboni, F., De Marchi, M., & Rauch, E. (2022). Development and evaluation of design guidelines for cognitive ergonomics in human-robot collaborative assembly systems. *Applied Ergonomics*, *104*, 103807. https://doi.org/10.1016/j.apergo.2022.103807

Gualtieri, L., Rauch, E., & Vidoni, R. (2021). Emerging research fields in safety and ergonomics in industrial collaborative robotics: A systematic literature review. *Robotics and Computer-Integrated Manufacturing*, *67*, 101998. https://doi.org/10.1016/j.rcim.2020.101998

Hall, E. (1966). *The hidden dimension* (Vol. 609). Doubleday. https://www.academia.edu/43785083/The_Hidden_Dimension_Edward_Hall

Hancock, G., Longo, L., Young, M., & Hancock, P. (2021). Mental workload. In *Handbook of human factors and ergonomics* (pp. 203–226). John Wiley & Sons, Inc. https://doi.org/10.1002/9781119636113.ch7

Hart, S. G. (1986). *NASA task load index (TLX): Computerized version-volume 1.0*. NASA Ames Research Center. Moffett Field, CA, United States. https://ntrs.nasa.gov/citations/20000021487

Hassenzahl, M., & Tractinsky, N. (2006). User experience - a research agenda. *Behaviour & Information Technology*, *25*(2), 91–97. https://doi.org/10.1080/01449290500330331

Hentout, A., Aouache, M., Maoudj, A., & Akli, I. (2019). Human–robot interaction in industrial collaborative robotics: A literature review of the decade 2008–2017. *Advanced Robotics*, *33*(15), 764–799. https://doi.org/10.1080/01691864.2019.1636714

Hoecherl, J., Schmargendorf, M., Wrede, B., & Schlegl, T. (2018). User-centered design of multimodal robot feedback for cobots of human-robot working cells in industrial production contexts. *ISR 2018; 50th International Symposium on Robotics*, 1–8. https://ieeexplore.ieee.org/abstract/document/8470632

Hornbæk, K., Kristensson, P. O., & Oulasvirta, A. (2025). Design practice. In K. Hornbæk, P. O. Kristensson, & A. Oulasvirta (Eds.), *Introduction to human-computer interaction*. Oxford University Press. https://doi.org/10.1093/oso/9780192864543.003.0032

Hsieh, H.-F., & Shannon, S. E. (2005). Three approaches to qualitative content analysis [ISBN: 1049-7323]. *Qualitative health research*, *15*(9), 1277–1288.

Hutchins, E., & Klausen, T. (1996). Distributed cognition in an airline cockpit. *Cognition and communication at work*, *1*, 15–34.

Ioniță, A.-R., Anghel, D.-C., & Boudouh, T. (2025). Mind, machine, and meaning: Cognitive ergonomics and adaptive interfaces in the age of industry 5.0. *Applied Sciences*, *15*(14), 7703. https://doi.org/10.3390/app15147703

Johnson, T., Tang, G., Fletcher, S. R., & Webb, P. (2016). Investigating the effects of signal light position on human workload and reaction time in human-robot collaboration tasks. In C. Schlick & S. Trzcieliński (Eds.), *Advances in ergonomics of manufacturing: Managing the enterprise of the future* (pp. 207–215). Springer International Publishing. https://doi.org/10.1007/978-3-319-41697-7_19

Karwowski, W., & Zhang, W. (2021). The discipline of human factors and ergonomics. In *Handbook of human factors and ergonomics* (pp. 1–37). John Wiley & Sons, Inc. https://doi.org/10.1002/9781119636113.ch1

Keshvarparast, A., Battini, D., Battaia, O., & Pirayesh, A. (2024). Collaborative robots in manufacturing and assembly systems: Literature review and future research agenda. *Journal of Intelligent Manufacturing*, *35*(5), 2065–2118. https://doi.org/10.1007/s10845-023-02137-w

Keyes, B., Micire, M., Drury, J. L., & Yanco, H. A. (2010). Improving human-robot interaction through interface evolution. In *Human-robot interaction*. IntechOpen. https://doi.org/10.5772/8140

Kluy, L., & Roesler, E. (2021). Working with industrial cobots: The influence of reliability and transparency on perception and trust. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *65*(1), 77–81. https://doi.org/10.1177/1071181321651110

Koay, K. L., Syrdal, D. S., Ashgari-Oskoei, M., Walters, M. L., & Dautenhahn, K. (2014). Social roles and baseline proxemic preferences for a domestic service robot. *International Journal of Social Robotics*, *6*(4), 469–488. https://doi.org/10.1007/s12369-014-0232-4

Koay, K. L., Syrdal, D. S., Walters, M. L., & Dautenhahn, K. (2007). Living with robots: Investigating the habituation effect in participants' preferences during a longitudinal human-robot interaction study [ISSN: 1944-9437]. *RO-MAN 2007 - The 16th IEEE International Symposium on Robot and Human Interactive Communication*, 564–569. https://doi.org/10.1109/ROMAN.2007.4415149

Legler, F., Trezl, J., Langer, D., Bernhagen, M., Dettmann, A., & Bullinger, A. C. (2023). Emotional experience in human–robot collaboration: Suitability of virtual reality scenarios to study interactions beyond safety restrictions. *Robotics*, *12*(6). https://doi.org/10.3390/robotics12060168

Lorenzini, M., Lagomarsino, M., Fortini, L., Gholami, S., & Ajoudani, A. (2023). Ergonomic human-robot collaboration in industry: A review. *Frontiers in Robotics and AI*, *9*. https://doi.org/10.3389/frobt.2022.813907

Luu, T., Nguyen, Q., Tran, T., Tran, M., Ding, S., Kua, J., & Hoang, T. (2025). Enhancing real-time robot teleoperation with immersive virtual reality in industrial IoT networks. *The International Journal of Advanced Manufacturing Technology*, *139*(11), 6233–6257. https://doi.org/10.1007/s00170-025-16236-w

Mara, M., Meyer, K., Heiml, M., Pichler, H., Haring, R., Krenn, B., Gross, S., Reiterer, B., & Layer-Wagner, T. (2021). CoBot studio VR: A virtual reality game environment for

transdisciplinary research on interpretability and trust in human-robot collaboration. https://openreview.net/forum?id=ZlAhl-wrYqH

Matheson, E., Minto, R., Zampieri, E. G. G., Faccio, M., & Rosati, G. (2019). Human-robot collaboration in manufacturing applications: A review. *Robotics*, *8*(4). https://doi.org/10.3390/robotics8040100

Mortimer, B. J. P., Elliott, L. R., & Miller, L. C. (2020). Designing tactile cues: Factors affecting perceived salience and recognition. *IEEE Transactions on Haptics*, *13*(4), 709–719. https://doi.org/10.1109/TOH.2020.2969668

Nelson, W. A. (2013). Design, research, and design research: Synergies and contradictions. *Educational Technology*, *53*(1), 3–11. https://www.jstor.org/stable/44430111

Nielsen, S., Jochum, E., Li, C., Chrysostomou, D., & Ordoñez, R. (2026). "robot emotions are not real!": Future factory workers' perceptions, attitudes, and experience of collaborative robots, conversational AIs, and AI-empowered, voice-enabled collaborative robots. *J. Hum.-Robot Interact.*, *15*(2), 40:1–40:39. https://doi.org/10.1145/3779295

Novakazi, F., Renoux, J., & Ellis. (2025). Prerequisites, perceptions, and performance: A mixed-methods study of assembly worker's expectations for human-robot collaboration [Under review, obtained with author permission]. *Human Factors and Ergonomics in Manufacturing & Service Industries*.

Oberauer, K. (2009). Chapter 2: Design for a working memory. In *Psychology of learning and motivation* (pp. 45–100, Vol. 51). Academic Press. https://doi.org/10.1016/S0079-7421(09)51002-X

Panagov, S. (2023). *Human centric collaborative workplace: The human robot interaction system perspective* [Doctoral thesis]. University of Basilicata. https://iris.unibas.it/handle/11563/165834#

Patil, S., Vasu, V., & Srinadh, K. V. S. (2023). Advances and perspectives in collaborative robotics: A review of key technologies and emerging trends. *Discover Mechanical Engineering*, *2*, 13. https://doi.org/10.1007/s44245-023-00021-8

Productive Robotics. (2023, August 14). *Why swiss productions, etogen precision & true precision chose the OB7 cobot*. https://www.youtube.com/watch?v=HrTFCaQdjqU

Productive Robotics. (2024, February 6). *7-axis blaze welding cobot with miller power source | high-speed robotic welding*. https://www.youtube.com/watch?v=5tOSAmMZQvU

Productive Robotics. (2025, June 14). *Parts marking made easy with the OB7 cobot | automation application showcase*. https://www.youtube.com/watch?v=SEqnzGDf3f4

Raider-Roth, M., Gold, M., Brydon-Miller, M., & Dorph, G. Z. (2021). Moving toward a utopian future one step at a time: Taking our future creating workshop online. *Journal of Participatory Research Methods*, *2*(1). https://doi.org/10.35844/001c.18689

Ramaraj, P., Sahay, S., Kumar, S. H., Lasecki, W. S., & Laird, J. E. (2019). Towards using transparency mechanisms to build better mental models. *Advances in cognitive systems: 7th goal reasoning workshop*, *7*, 1–6.

Reid, G. B., & Nygren, T. E. (1988). The subjective workload assessment technique: A scaling procedure for measuring mental workload. In *Advances in psychology* (pp. 185–218, Vol. 52). North-Holland. https://doi.org/10.1016/S0166-4115(08)62387-0

Riener, K., Patel, V., Jiang, L., Elahi, B., Suresh, M., Papa, L., & Luo, Y. (2025). Human-robot communication: Utilizing light-based signals to convey robot operating state. *Human Factors in Robots, Drones and Unmanned Systems*, *179*, 135–144. https://doi.org/10.54941/ahfe1006382

Rosén, J. (2021). Expectations in human-robot interaction. In H. Ayaz, U. Asgher, & L. Paletta (Eds.), *Advances in neuroergonomics and cognitive engineering* (pp. 98–105, Vol. 259). Springer International Publishing. https://doi.org/10.1007/978-3-030-80285-1_12

Salmon, P. M., Stanton, N. A., Walker, G. H., Baber, C., Jenkins, D. P., McMaster, R., & Young, M. S. (2008). What really is going on? review of situation awareness models for individuals and teams. *Theoretical Issues in Ergonomics Science*, *9*(4), 297–323. https://doi.org/10.1080/14639220701561775

Satisfying Tech. (2025, October 30). *Amazing bulldozer mass production process! impressive modern factory*. Retrieved April 21, 2026, from https://www.youtube.com/watch?v=NC7vEpnW0xo

Sauer, V., Sauer, A., & Mertens, A. (2021). Zoomorphic gestures for communicating cobot states. *IEEE Robotics and Automation Letters*, *6*(2), 2179–2185. https://doi.org/10.1109/LRA.2021.3060416

Scholz, C., Cao, H.-L., El Makrini, I., Niehaus, S., Kaufmann, M., Cheyns, D., Roshandel, N., Burkiewicz, A., Shhaitly, M., Imrith, E., Genoe, J., Rottenberg, X., Gerets, P., & Vanderborght, B. (2025). Improving robot-to-human communication using flexible display technology as a robotic-skin-interface: A co-design study. *International Journal of Intelligent Robotics and Applications*, *9*(1), 146–163. https://doi.org/10.1007/s41315-024-00343-0

Schrepp, M. (2023). *User experience questionnaire handbook* (Version 11). https://www.ueq-online.org/

Schuler, D., & Namioka, A. (Eds.). (1993). *Participatory design: Principles and practices*. CRC Press. https://doi.org/10.1201/9780203744338

Shah, R., Doss, A. S. A., & Lakshmaiya, N. (2025). Advancements in AI-enhanced collaborative robotics: Towards safer, smarter, and human-centric industrial automation. *Results in Engineering*, *27*, 105704. https://doi.org/10.1016/j.rineng.2025.105704

Shantui Construction Machinery Co., Ltd. (2026). *About shantui*. Retrieved April 21, 2026, from https://www.shantui-global.com/about/about-us.htm

Shourangiz, E., Ghafari, F., & Wang, C. (2025). Human-robot collaboration integrated with virtual reality in construction and manufacturing industries: A systematic review. *Virtual Reality & Intelligent Hardware*, *7*(4), 317–343. https://doi.org/10.1016/j.vrih.2024.08.004

St. Clair, A., & Mataric, M. (2015). How robot verbal feedback can improve team performance in human-robot task collaborations. *Proceedings of the Tenth Annual ACM/IEEE International Conference on Human-Robot Interaction*, 213–220. https://doi.org/10.1145/2696454.2696491

Stanton, N. A., Salmon, P. M., Walker, G. H., & Jenkins, D. P. (2010). Is situation awareness all in the mind? *Theoretical Issues in Ergonomics Science*, *11*(1), 29–40. https://doi.org/10.1080/14639220903009938

Takayama, L., Dooley, D., & Ju, W. (2011). Expressing thought: Improving robot readability with animation principles. *Proceedings of the 6th international conference on Human-robot interaction*, 69–76. https://doi.org/10.1145/1957656.1957674

Tang, G., Webb, P., & Thrower, J. (2019). The development and evaluation of robot light skin: A novel robot signalling system to improve communication in industrial human–robot collaboration. *Robotics and Computer-Integrated Manufacturing*, *56*, 85–94. https://doi.org/10.1016/j.rcim.2018.08.005

UEQ Team. (2024). *User experience questionnaire*. https://www.ueq-online.org/

Universal Robots. (2022, May 6). *Top five industrial applications of cobots*. Retrieved April 23, 2026, from https://www.universal-robots.com/blog/top-five-industrial-applications-of-cobots/

Universal Robots. (2025, October 7). *Dozens of robots support the production of millions of sensor-rich units*. https://www.universal-robots.com/case-stories/sensata-technologies/

Varela, F. J., Rosch, E., & Thompson, E. (1992). *The embodied mind: Cognitive science and human experience*. MIT Press.

Vermeulen, J., Caldwell, G., Teixeira, M., Burden, A., & Guertler, M. (2024). To safety and beyond! a scoping review of human factors enriching the design of human-robot collaboration. *Interaction Design and Architecture(s)*, (61), 42–65. https://doi.org/10.55612/s-5002-061-001

Villani, V., Fenech, G., Fabbricatore, M., & Secchi, C. (2023). Wrist vibration feedback to improve operator awareness in collaborative robotics. *Journal of Intelligent & Robotic Systems*, *109*(3), 45. https://doi.org/10.1007/s10846-023-01974-4

Villani, V., Pini, F., Leali, F., & Secchi, C. (2018). Survey on human–robot collaboration in industrial settings: Safety, intuitive interfaces and applications. *Mechatronics*, *55*, 248–266. https://doi.org/10.1016/j.mechatronics.2018.02.009

Weiss, A., Wortmeier, A.-K., & Kubicek, B. (2021). Cobots in industry 4.0: A roadmap for future practice studies on human–robot collaboration. *IEEE Transactions on Human-Machine Systems*, *51*(4), 335–345. https://doi.org/10.1109/THMS.2021.3092684

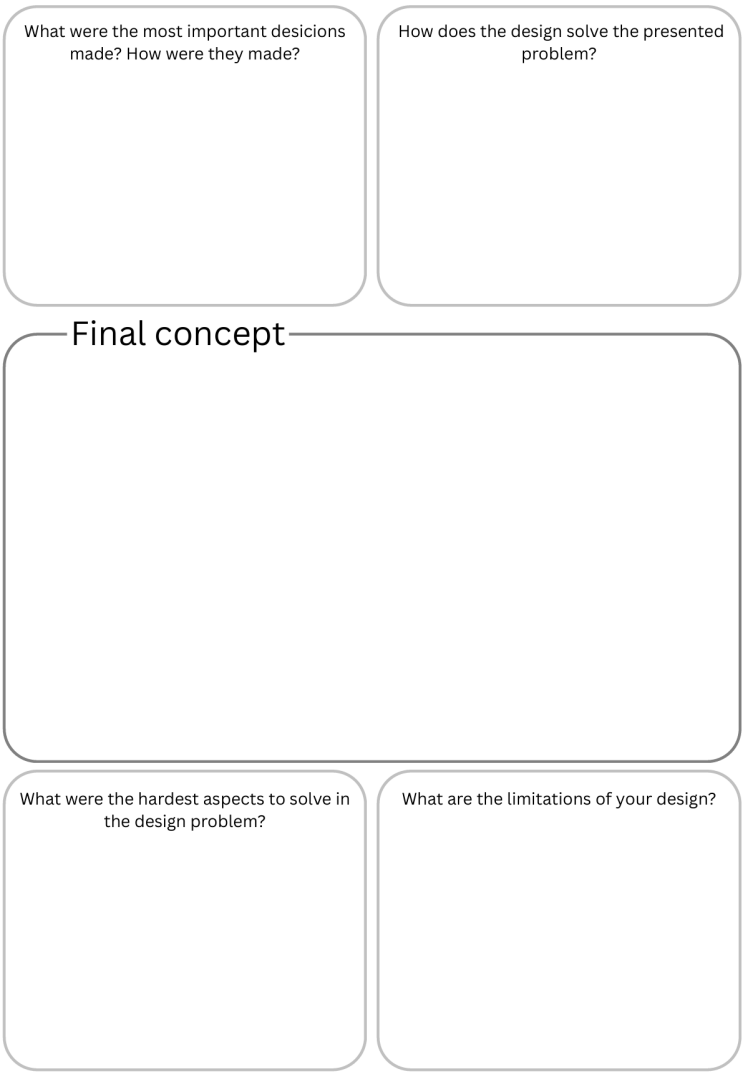
Wickens, C. D. (2008). Multiple resources and mental workload. *Human Factors*, *50*(3), 449–455. https://doi.org/10.1518/001872008X288394

Wickens, C. D., & Carswell, C. M. (2021). Information processing. In *Handbook of human factors and ergonomics* (pp. 114–158). John Wiley & Sons, Inc. https://doi.org/10.1002/9781119636113.ch5

Wogalter, M. S., Mayhorn, C. B., & Laughery Sr., K. R. (2021). Warnings and hazard communications. In *Handbook of human factors and ergonomics* (pp. 644–667). John Wiley & Sons, Inc. https://doi.org/10.1002/9781119636113.ch24

11 Appendix

11.1 A3 Final Design Template



11.2 Workshop Consent Form

Project: Human-robot communication in factory settings
Örebro University

Consent Form

I _____ voluntarily agree to participate in this research study.

1. I understand that even if I agree to participate now, I can withdraw at any time without any consequences of any kind.
2. I have had the purpose and nature of the study explained to me and I have had the opportunity to ask questions about the study.
3. I understand that the session will **not** be recorded using audio or video, but the researcher(s) conducting the session will take notes on paper.
4. I understand that access to the interview notes will be limited to the researchers involved in the project.
5. I understand that the designs I will contribute to create during the workshop, as well as the notes taken by researchers, may be used in future scientific publications.
6. I understand that my identity will remain **anonymous** in any report on the results of this research and that **no personal information** will be used.
7. I understand that the researchers conducting the study will keep the signed consent forms and any material resulting from the workshop until two years after the research project has ended.
8. I understand that I am entitled to access the information I have contributed to create at any time while it is stored as specified above.
9. I understand that I am free to contact any of the people involved in the research to seek further clarification and information.
10. I do not expect to receive any benefit or payment for my participation.

Researchers supervising the study:

- Jennifer Renoux, Senior Researcher in Computer Science, Örebro University, Sweden. E-mail:jennifer.renoux@oru.se

_____ _____
Signature of the participant Date

_____ _____
Signature of the supervising researcher Date

11.3 Descriptions for Refined Designs

Design 1 – Swedish

Medarbetaren arbetar enskilt på sin uppgift på bordet bredvid cobotten, men medarbetaren kan behöva ingripa hos cobotten ibland. Cobotten ger därför signaler baserat på sin status, genom rörelse och ljussignaler från sin basplatta enligt nedan:

1. Fungerande – Cobotten jobbar på och behöver ingen mänskligt ingripande. Ingen signal.

2. Färdig – Cobotten är klar med sin uppgift och behöver mänskligt ingripande för att fortsätta. Exempelvis en ny uppgift eller nya obearbetade komponenter att arbeta på. Den sjunker ihop och basplattan lyser grönt medan den väntar.

3. Behöver assistans – Själva cobotten fungerar, men något annat har gjort att den inte kan fortsätta med sin uppgift och mänskligt ingripande krävs. Exempelvis att behöva hjälp med att få loss en komponent eller behöva leverans av något material som är nödvändigt för uppgiften men som tagit slut. Den ställer sig rakt upp för att tydligt signalera att den ej arbetar och basplattan lyser gult.

4. Tekniskt fel – Själva cobotten har ett teknist fel och behöver mänskligt ingripande i form av reparation. Exempelvis har en av lederna slutat fungerat, eller griparen sitter löst. Den ställer sig rakt upp (om möjlligt) för att tydligt signalera att den ej arbetar och basplattan lyser rött. Ifall inget ingripande sker på en stund medan antingen status 3 eller 4 är aktiv så börjar cobotten vinka i intervaller för att fånga uppmärksamhet (ifall rörelse fortfarande är möjligt). När ingripande väl sker så kan arbetaren hitta mer information om vad som ska göras på en skärm nära cobotten.

Design 1 – English

The coworker is working individually on their task at the table next to the cobot, but the worker may need to occationally intervene at the cobot. The cobot therfore produces signals based on its status, through movement and light from its baseplate as follows:

1. Working – The cobot is working away and does not need any human intervention. No signal.

2. Finished – the cobot is done with its task and needs human intervention to continue. For instance, a new task or new unprocessed components to work on. It slumps down and the baseplate lights up green while it waits.

3. Needs assistance – the cobot itself is functioning, but something else has gotten in the way of its task and human intervention is needed. For instance, needing help to dislodge a stuck component, or need a delivery of some necessary material that has run out. It also extends upward to indicate clearly that it is not working and the baseplate lights up yellow.

4. Technical error – The cobot itself has a malfunction and requires human intervention in the form of repairs. For example, one of the joints stopped working, or the gripper has come loose. It extends upward (if possible)to indicate clearly that it is not working and the baseplate lights up red.

If no intervention happens for a while when point 3 or 4 is active, the cobot will start waving in intervals to get the workers attention (if movement is still possible). When the worker does intervene, they can find more information on what needs to be done on a nearby screen.

Design 2 – Swedish

En medarbetare arbetar med sin egen uppgift, men går ibland i närheten av cobotten. Ifall cobotten behöver något från medarbetaren så säger den det när den detekterar att medarbetaren är i närheten. Mer specifik information kring vad cobotten behöver hjälp med finns på en skärm i cobottens närhet. Exempelvis angående tekniska fel på själva cobotten eller andra problem som hindrar uppgiften (kanske någonting är i vägen eller har fastnat).

Design 2 – English

A coworker is working on their own task, but sometimes it walks near the cobot. In case the cobot needs something from the worker it states what it needs when it can detect that the worker is nearby. More specific information about what the cobot needs assistance with can be found on a screen near the cobot. It could for instance be malfunctions with the cobot itself, or other problems preventing the cobot from doing its task (perhaps something is in the way or stuck).

Design 3 – Swedish

En medarbetare och en cobot utför olika uppgifter, men samtidigt och på samma komponent. För att medarbetaren ska kunna förutspå cobottens rörelser så utför cobotten förutsägbara rörelser som i kolumn ett i bilden ovan. Cobotten har lämnat den svarta delen till vänster, och ska nu plocka upp den orangea delen längst till höger. Cobotten gör en stor och tydlig rörelse samt pekar sin gripare tidigt mot delen som ska plockas. Det gör att det blir tydligt tidigt i rörelsen att cobotten sträcker sig efter den orangea delen snarare än den gröna som ligger innan. Detta står i kontrast till kolumn två som visar hur cobotten inte bör röra sig, trots att det är den mest direkta vägen till delen som ska plockas. I kolumn två är inte lika lätt för arbetaren att förutsäga tidigt vilken del som ska plockas och detta är fortfarande tvetydigt i det tredje rörelsemomentet.

Design 3 – English

A coworker and a cobot are doing different tasks, but at the same time and on the same component. In order for the worker to predict the cobots motions, the cobot moves predictably as seen in column one above. The cobot has just placed the black part to the left, and will now go pick up the orange part to the far right. The cobot performs a large and clear movement and points its gripper towards the part that is to be picked up. This means it is clear early in the movement that the cobot is reaching for the orange part as opposed to the green part that comes before. This is in contrast to column two, which shows how the cobot should not move, even if it would be the most direct way to the part that is to be picked up. In column two it is not as easy for the worker to predict which part will be picked up early in the movement, and this is still ambiguous even in the third frame.

Design 4 – Swedish

En medarbetare och en cobot utför olika uppgifter, men samtidigt och på samma komponent. För att indikera förloppet av cobottens uppgift finns inbyggda lampor vid cobottens nedre del, dessa signalerar både förloppet men också att cobotten därför är i aktiv rörelse. Lamporna indikerar även när uppgiften är färdig (grönt ljus), och ifall det uppkommer något fel eller cobotten behöver assistans (rött ljus) (detaljer kan sedan ses på en skärm). Längst ut på cobottens arm finns en ring med lampor, lampan som lyser blått indikerar riktningen som roboten kommer röra sig.

När du svarar på frågorna nedan, tänk på båda aspekterna av designen: statuslamporna på nedre delen och ringen med lampor längst ute på armen.

Design 4 – English

A coworker and a cobot are doing different tasks, but at the same time and on the same component. To indicate progress of the cobots task, there are built-in lights at the lower part of the cobot, these signal both the progress but also that the cobot therefore is in active movement. The lights also indicate when the task is completed (green light), and in case there is an error or the cobot needs assistance (red light) (details can then be found on a screen). At the end of the cobot arm there is a ring of lights, the blue light that is lit indicates the direction of the cobots upcoming movement.

When answering the questions below, make sure to consider both the progress bar and the ring of lights aspects of the design.

Design 5 – Swedish

1. En medarbetare och en cobot utför olika uppgifter, men samtidigt och på samma komponent. Cobotten är utrustad med en töjbar skärm som sitter hela vägen runt en del av armen. Skärmen kan innehålla olika typer av information. Exempelvis:
2. Förloppet på den nuvarande uppgiften, samt den uppgift som kommer härnäst och hur lång tid det uppskattas att den uppgiften kommer ta.
3. Blått ljus som skiner åt det håll som cobotten kommer röra sig. Indikerar även att cobotten är i aktiv rörelse.
4. Statusindikator som till exempel kan visa att det är ett tekniskt fel och vad det felet gäller. Skulle även kunna indikera andra typer av status, exempelvis om cobotten behöver annan assistans eller om dess säkerhetssystem uppfattar att en medarbetare är i närheten.
5. Instruktioner som visar uppgiften samt vilket steg av uppgiften som just nu utförs. Detta skulle kunna vara cobottens uppgift, men kan även vara medarbetarens uppgift för att stötta medarbetarens arbete.

Skärmen skulle vara anpassningsbar i relation till vilken medarbetare som samarbetar med cobotten eftersom olika medarbetare kanske vill att cobotten visar olika typer av information. När du svarar på frågorna nedan, tänk på alla olika exempel på information som skärmen kan ha.

Design 5 – English

A coworker and a cobot are doing different tasks, but at the same time and on the same component. The cobot is equipped with a bendable screen that spans all the way around one part of the arm. The screen could contain various types of information. For example:

1. The progress on the current task, as well as the upcoming task and how long it is estimated that task will take.
2. Blue light that shines in the direction of the cobots upcoming movement. Also indicates that the cobot is in active movement.
3. Status indicator that, for instance, could show a technical error notification and what the error is. It could also indicate other types of status, for instance if the cobot needs another type of assistance or if its safety system is detecting that a worker is nearby.
4. Instructions that show the task, and which step is currently being executed. This could be the cobots task, but it could also be to show the workers task to support their work.

The screen would be adaptable in relation to which worker is working with the cobot since different workers may wish the cobot to display different types of information. When answering the questions below, make sure to consider each example of information that can be displayed.

11.4 Expert Evaluation Consent Form

1. I voluntarily agree to participate in this research study.
2. I understand that even if I agree to participate now, I can withdraw at any time without consequences of any kind by simply closing the questionnaire window.
3. I have had the purpose and nature of the study explained to me and I have had the opportunity to ask questions.
4. I understand that access to the responses I provide in this study is restricted to researchers involved in the project.
5. I understand that the responses I provide in this study may be used in future scientific publications.
6. I understand that my identity will remain anonymous and that no personal information will be gathered.
7. I understand that the responses I provide in this study may be kept until two years after the research project has ended.
8. I understand that I am free to contact any of the people involved in the research to seek further clarification and information.
9. I do not expect to receive any benefit or payment for my participation.

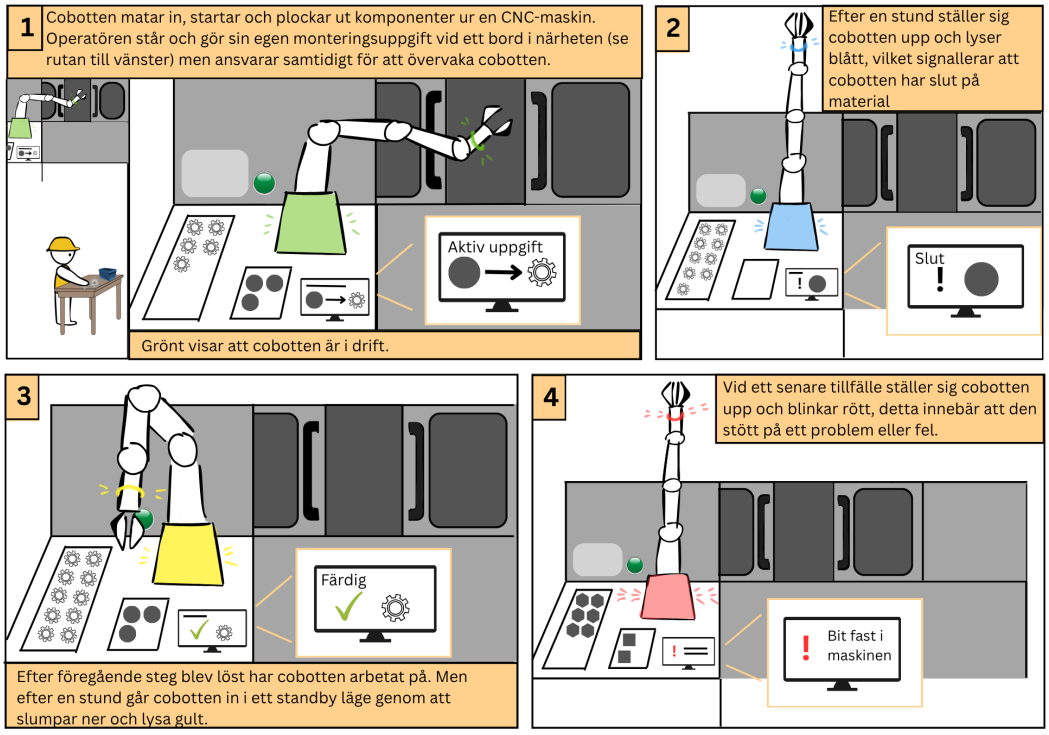
I have read and consent to the above: Yes / No (exit form)

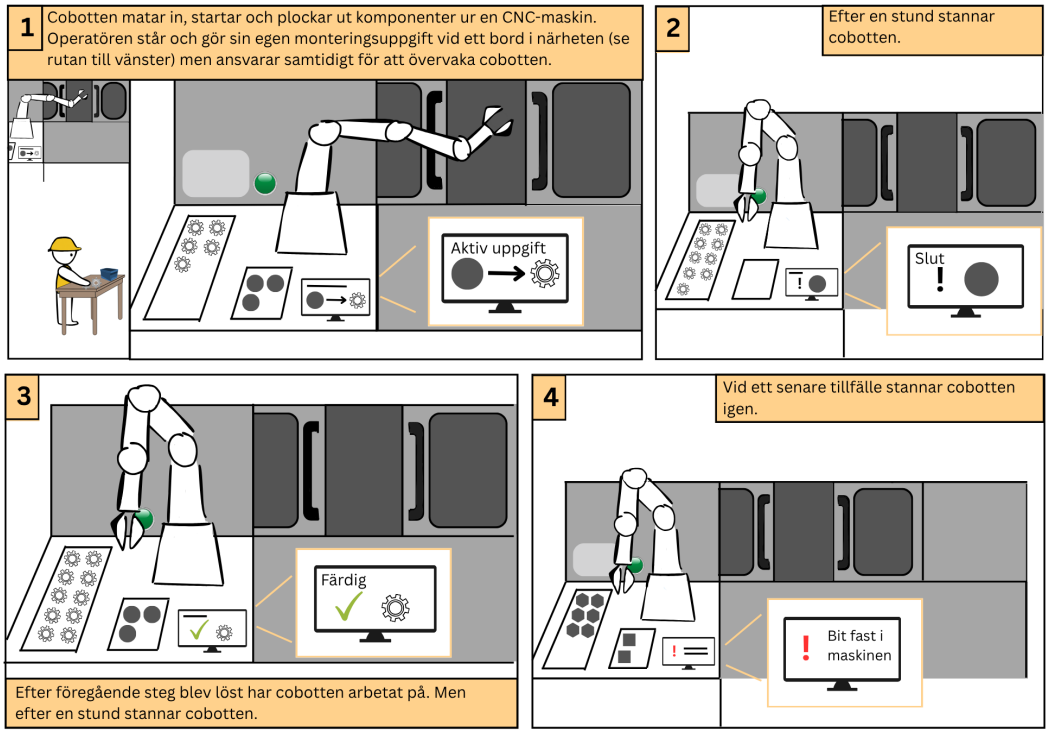
11.5 Empirical Assessment Consent Form

1. Jag deltar frivilligt i denna studie.
2. Jag förstår att även om jag samtycker till deltagande nu, så kan jag dra mig ur när som helst helt utan konsekvenser genom att bara stänga ner enkätfönstret/lämna.
3. Jag har fått studiens syfte och natur förklarad (se ovan).
4. Jag förstår att tillgång till svaren jag ger i denna studie är begränsad till forskare inom projektet.
5. Jag förstår att svaren jag ger i denna studie kan komma att bli del av framtida vetenskapliga publikationer.
6. Jag förstår att min identitet förblir anonym och att ingen personlig information kommer samlas.
7. Jag förstår att jag kan kontakta forskarna inblandade för förtydligande och information.

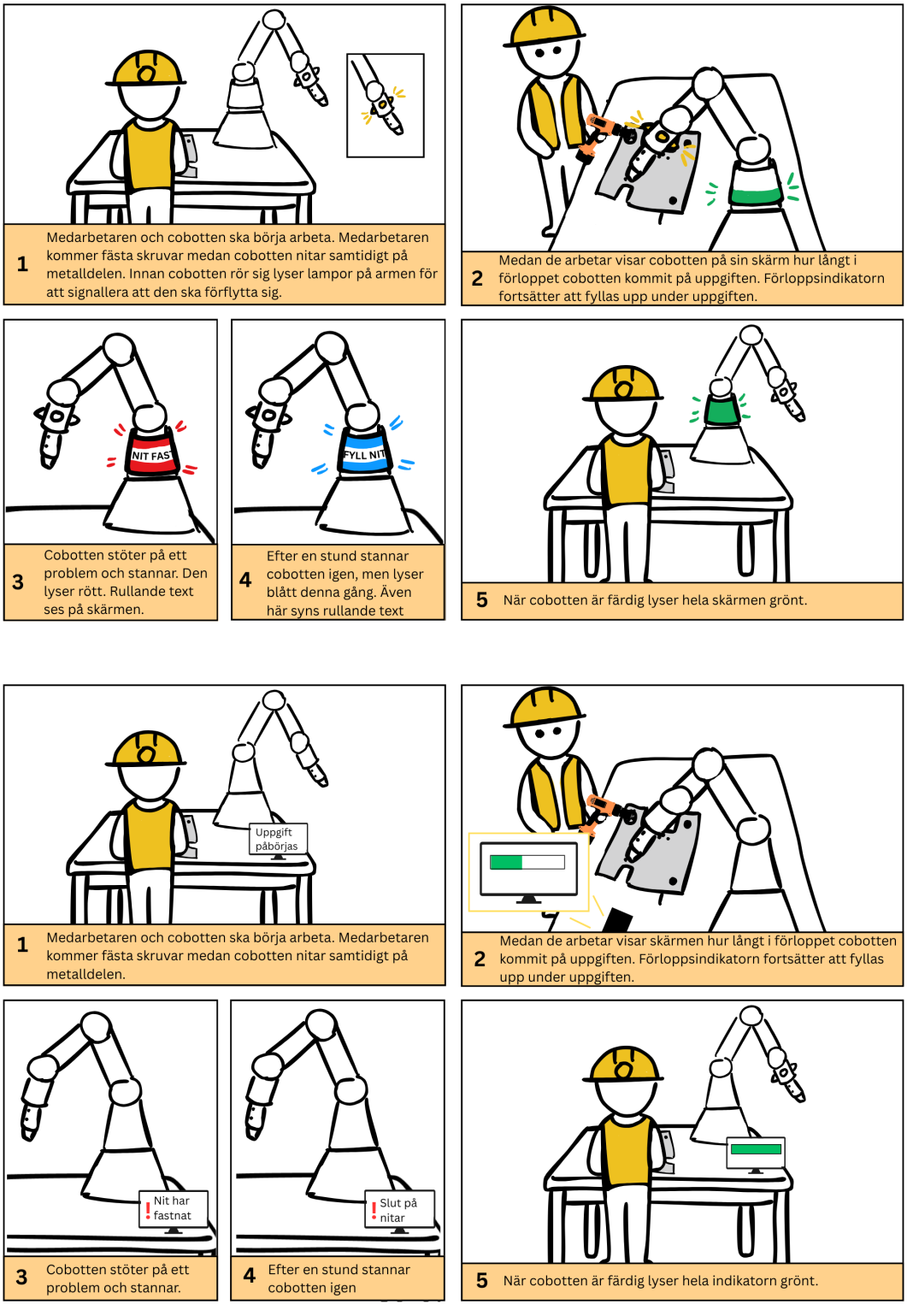
Jag har läst och samtycker till det ovanstående (ruta att markera)

11.6 Independent Storyboards Swedish





11.7 Simultaneous Storyboards Swedish



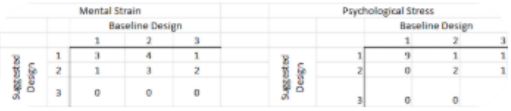
11.8 Generative AI Use

Generative AI (GPT-5.5 Instant) has been used to polish language and check consistency. Below is an example of a starting prompt used:

"While I have read through my thesis for basic language issues some things could benefit from being double checked, but not nit picked. I want the focus to be on:

- Spelling errors/typos
- Grammar issues (e.g incorrect pluralisations or possessives, punctuation, NOT "general" improvements unless absolutely necessary for clarity and readability)
- Cohesion and a "red thread" (highlight any particular issues. One I am aware of is that the very first paragraph of the theory section is not introducing the section very well, I'm considering remeding this by adding a section introduction explaining what content the section will contain. If such a introduction section is needed elsewhere, this can be pointed out)
- Acronyms (ensuring they are written out at the first occurance and that they are consistently used. The main ones are RtH, HRC,HRI, and UX, be weary of other ones)
- Writing numerals with numbers or letters. This is currently inconsistent, I have tried to spell out anything 0-9 as per APA (but of course not in the results section for the statistics), but may have missed some (though "design 1, 2, 3" etc should use numbers.)
- Any unreadable sentences that break in logic or are particularly long or confusing.
- Refering to other sections should include the section number
- Condistency in general, word use, captions, headings.
- Overall matter-of-factly language, and mainly using an active voice. Point out particularly casual formulations."

It has also been used as a help to transfer the thesis into LaTeX, e.g. code for tables/figures or learning how functions work (as the author is new to LaTeX). Tables/figures plus legends were always first created by the author and Generative AI was only an aid in displaying them using LaTeX (including aiding with further edits/troubleshooting of LaTeX code). The author ensured that numbers had been retained correctly and wrote table/figure captions. An example of the start of a table conversion can be seen below:



Mental Strain

Suggested Design	Baseline Design 1	2	3
1	3	4	1
2	1	3	2
3	0	0	0

Psychological Stress

Suggested Design	Baseline Design 1	2	3
1	9	1	1
2	0	2	1
3	0	0	

Can you make these contingency tables in LaTex, side by side. I have a preamble I just need the tables.

You can place them side-by-side using `minipage` and a standard `tabular`. Here's a clean version that reproduces the tables shown:

LaTeX

```
\begin{table}[ht]
\centering

\begin{minipage}{0.45\textwidth}
\centering
\caption{Mental Strain}
\[
\begin{array}{c|ccc}
 & \multicolumn{3}{c}{\text{Baseline Design}} \\
\text{Suggested Design} & 1 & 2 & 3 \\
\hline
1 & 3 & 4 & 1 \\
2 & 1 & 3 & 2 \\
3 & 0 & 0 & 0
\end{array}
\]
\end{minipage}
\hfill
\begin{minipage}{0.45\textwidth}
\centering
\caption{Psychological Stress}
\[
\begin{array}{c|ccc}
 & \multicolumn{3}{c}{\text{Baseline Design}} \\
\text{Suggested Design} & 1 & 2 & 3 \\
\hline
1 & 9 & 1 & 1 \\
2 & 0 & 2 & 1 \\
3 & 0 & 0 & 0
\end{array}
\]
\end{minipage}

\end{table}
```

If you want the row label "Suggested Design" rotated vertically as in your screenshot, I can also provide a version using the `rotating` or `graphicx` package.

Yes. I need the suggested design to be rotated. Also I want them not to be as two seperate tables. Just one table caption, but two parts to the table with the headings of mental strain and psychological stress.